A Project Report on

SpotNFix: Smart Pothole Reporting & Severity Detection System

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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This is to certify that the project entitled "**SpotNFix: Smart Pothole Reporting & Severity Detection System**" is a bonafide work of **Aditya Sawant (22106084), Jay Yadav (23206007), Pranal Vernekar (23206008), Manas Jagtap (23206011)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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Abstract

Road infrastructure maintenance remains a significant challenge in rapidly urbanizing regions, where potholes contribute to road accidents, vehicle damage, and traffic inefficiencies. Existing reporting mechanisms largely depend on manual complaints through helplines or municipal portals, which often lack proper validation, severity assessment, and transparency in repair workflows. These limitations lead to delayed responses, inefficient prioritization, and reduced accountability among stakeholders.

To address these challenges, this paper presents SpotNFix, an AI-driven, web-based pothole reporting and management system. The platform enables citizens to upload images of potholes, which are automatically geotagged using location data. A deep learning–based object detection model verifies whether the uploaded image contains a pothole, reducing false or duplicate reports. Additionally, the system performs severity estimation to classify potholes into different levels, allowing authorities to prioritize critical repairs effectively.

SpotNFix also introduces a transparent bidding mechanism, where contractors can compete for repair tasks, ensuring cost efficiency and accountability. Role-based dashboards for citizens, government officials, and contractors facilitate seamless communication and workflow tracking. By integrating machine learning, geolocation services, and real-time analytics, the system transforms traditional reactive maintenance into a proactive, data-driven approach, offering a scalable and efficient solution for smarter urban road infrastructure management

Keywords—Pothole Detection, Intelligent Transportation Systems, Smart Cities, Deep Learning, YOLO, Infrastructure Monitoring

Contents

1. Introduction 1
2. Literature Survey/ Existing System 3
3. Limitation of Existing System 8
4. Problem Statement and Objective 10
5. Proposed System 13
5.1 System Overview 14
5.2 Citizen Module Citizens act as primary data contributors 14
5.3 Government Officer Module 14
5.4 Contractor Module 14
5.5 AI-Based Verification and Automation 15
5.6 Transparency and Community Engagement 15
5.7 Pothole Lifecycle 15
5.8 Dataflow Diagram 16
5.9 Use Case Diagram 18
5.10 Case Study and Methodology 18
5.10.1 Case Context 18
5.10.2 Methodological Framework 19
5.10.3 YOLO-Based Pothole Detection Methodology 19
5.10.4 MobileNetV2-Based Severity Classification 20
6. Experimental Setup 22
6.1 Dataset Details 23
6.1.1 Dataset for Pothole Detection (YOLO Model) 23
6.1.2 Dataset for Severity Classification (MobileNet CNN) 23
6.2 Performance Evaluation Parameters (Validation Metrics) 24
6.3 Software and Hardware Setup 24
6.3.1 Software Setup 24
6.3.2 Hardware Setup 25
6.4 Implementation 26
7. Results and Discussion 30
7.1 System-Level Improvements 30
7.1.1 Reduction in Manual Effort through System Automation 31
7.1.2 Analysis of Geolocation Accuracy and Positional Variations 32
7.1.3 Comparative Cost Analysis – Traditional Estimation vs Bidding Mechanism 32

7.1.4 Reduction in time through System Automation across pothole lifecycle 33
7.2 Model Metrics (YOLO) 34
7.3 Evaluation Metrics (MobileNet) 36
8. Conclusion and Future Work 39
References 41
Appendix 43
Publication 46
Acknowledgements 47

LIST OF FIGURES

Figure 5.1 : Architecture of the Proposed System 13
Figure 5.7.1 : Pothole Lifecycle 15
Figure 5.8.1 : Data Flow Diagram Level-0 16
Figure 5.8.2 : Data Flow Diagram Level-1 16
Figure 5.8.3 : Data Flow Diagram Level-2 17
Figure 5.9.1 : Use Case Diagram 18
Figure 6.4.1 : Contractor Dashboard Interface 26
Figure 6.4.2 : Pothole Detection & Upload Interface 26
Figure 6.4.3 : Live Map View with Pothole Details 27
Figure 6.4.4 : Contractor Bidding Interface 27
Figure 6.4.5 : Bidding Details Interface 28
Figure 6.4.6 : Approve Pothole Interface 28
Figure 6.4.7 : Community Discovery Interface 29
Figure 6.4.8 : Community Management Interface 29
Figure 7.2.1 : F1-Confidence Curve showing optimal threshold around 0.35 34
Figure 7.2.2 : Precision-Confidence Curve demonstrating increasing precision at higher threshold 35
Figure 7.2.3 : Precision-Recall Curve showing mAP@0.5 of 0.773 35
Figure 7.2.4 : Confusion Matrix showing true positives, false positives and false negatives 36
Figure 7.3.1 : Precision, Recall & F1 per class for MobileNetV2 Classification 37
Figure 7.3.2 : ROC curves and AUC scores for MobileNetV2 Classification 37
Figure 7.3.3 : Confusion matrix for MobileNetV2 severity classification 38

LIST OF TABLES

Table 2.1 : Literature Survey of the previous studies 6
Table 7.1.1 : Comparison of Manual vs Automated Pothole Management Processes.....31
Table 7.1.2 : Geolocation Accuracy Analysis of Reported Pothole Analysis 32
Table 7.1.3 : Comparative Cost Analysis of Traditional Estimation and Bidding Mechanism 32
Table 7.1.4 : Time Reduction Across Pothole Management Lifecycle Using Automation 33
Table 7.3.1 : MobileNetV2 Severity Classification Metrics.....37

ABBREVIATION

AI	*Artificial Intelligence*
ML	*Machine Learning*
CNN	*Convolutional Neural Network*
YOLO	Y*ou Only Look Once*
ONNX	*Open Neural Network Exchange*
API	*Application Programming Interface*
GPS	*Global Positioning System*
mAP	*Mean Average Precision*
ROC	*Receiver Operating Characteristic*
AUC	*Area Under the Curve*
Adam	*Adaptive Moment Estimation*
SDG	*Sustainable Development Goals*
AMP	*Automatic Mixed Precision*
SQL	*Structured Query Language*
IoT	*Internet of Things*
GPU	*Graphics Processing Unit*
GSM	*Global System for Mobile Communications*
IMU	*Inertial Measurement Unit*
HSV	*Hue, Saturation, Value*
EXIF	Exchangeable Image File Format
SVM	*Support Vector Machine*

Chapter 1

Introduction

Road infrastructure maintenance in rapidly urbanizing cities is under increasing pressure due to population growth, rising vehicle density, and environmental factors such as heavy rainfall and temperature variations. These challenges lead to the frequent formation of potholes, which significantly impact road safety, traffic efficiency, and vehicle durability. Cities like Mumbai, characterized by high population density and intense monsoon seasons, experience accelerated road degradation, making timely detection and repair critical. Traditional pothole reporting mechanisms rely on manual complaints through helplines or municipal portals, which often lack proper validation, prioritization, and transparency in execution. As a result, delays in repair, duplicate reports, and inefficient resource allocation remain persistent issues.

In urban environments, the absence of reliable verification systems leads to inaccurate or false reporting, making it difficult for authorities to make informed decisions. Manual inspection processes are time-consuming and fail to scale with the growing number of complaints. Existing approaches in research primarily focus on pothole detection using computer vision techniques such as YOLO-based object detection models and convolutional neural networks, as well as sensor-based detection using smartphone accelerometers. While these methods improve detection accuracy, they often lack integration with a complete repair workflow, including prioritization, contractor management, and accountability. Furthermore, most

systems do not provide severity estimation, which is essential for efficient maintenance planning.

This study presents SpotNFix, an AI-driven, web-based pothole reporting and management system that integrates machine learning-based detection with a transparent and structured repair workflow. The system enables citizens to upload geotagged images of potholes, which are automatically verified using a YOLO-based object detection model. Severity classification is performed using AI models to prioritize repairs, ensuring that high-risk potholes are addressed first. The platform incorporates a role-based ecosystem involving citizens, government authorities, and contractors, supported by features such as live mapping, real-time dashboards, and a competitive bidding mechanism for repair tasks. Designed for scalability and real-time deployment, SpotNFix enhances transparency, reduces manual workload, and transforms traditional reactive maintenance into a proactive, data-driven infrastructure management system.

Chapter 2

Literature Survey/ Existing System

Existing research on pothole detection and road damage monitoring demonstrates that automated and intelligent systems are increasingly viewed as essential for improving road safety and infrastructure maintenance. A substantial body of work has focused on leveraging machine learning and deep learning techniques to identify potholes using images, videos, or sensor data collected from everyday devices [1], [2]. These studies emphasize real-time detection, cost efficiency, and scalability as critical requirements for practical deployment.

Several studies employ deep neural networks for image-based pothole detection. Prior work on real-time pothole detection has utilized YOLO-based architectures trained on images and video streams captured from dashboard cameras, mobile phones, and embedded devices such as NVIDIA Jetson platforms [3], [4]. These approaches demonstrate that object detection frameworks can accurately localize potholes under real-world conditions. Extensions and variations on the object detection paradigm, including improved backbone and neck modules and dilated convolutional layers, report improved performance in detecting irregularly shaped or partially occluded road defects [5].

Some systems further integrate GPS and GSM modules to geotag detected potholes and automatically notify authorities, highlighting the potential for low-cost, automated reporting mechanisms [6]. However, such systems are largely limited to detection and basic notification, offering minimal support for repair prioritization or long-term infrastructure management.

Related research has extended object detection to broader road damage classification, addressing multiple forms of surface deterioration such as cracks, corrugation, and surface depressions [7].

More recent YOLO-based studies and comparative detection papers examine trade-offs between model size, inference speed, and detection accuracy to enable deployment on resource-constrained platforms [3], [8]. These systems often incorporate real-time geolocation and web-based reporting platforms to encourage public participation. While this improves detection coverage and citizen involvement, most approaches remain focused on identification rather than coordinated repair workflows, contractor accountability, or transparent decision-making processes.

In contrast, another stream of literature explores sensor-based pothole detection using smartphones. By leveraging built-in accelerometers and gyroscopes, researchers propose low-cost solutions that analyze vibration signatures to classify road events as normal, pothole, or bump [1], [2]. Deep learning models, including transfer-learning approaches with architectures such as Inception V3, have demonstrated high classification accuracy on sensor datasets. Although these methods scale well through crowdsensing, they lack visual verification and provide limited information on pothole size, severity, and spatial characteristics, which are essential for prioritization and repair planning.

Additional studies focus on deep learning–based image classification rather than full object detection. Classification-based methods categorize road images into binary or multiclass labels such as “pothole,” “crack,” and “no damage” using convolutional neural networks [9]. While effective for basic identification, these approaches do not offer precise localization or severity metrics, limiting their usefulness for maintenance scheduling and resource allocation. Complementary work in hybrid ML pipelines integrates classical feature engineering with deep models to improve robustness across diverse road surfaces and lighting conditions [8], but challenges remain in translating detections into actionable maintenance plans.

Comparative analyses of machine learning techniques further illustrate trade-offs between accuracy, computational efficiency, and deployment feasibility [1], [3], [8]. Using data collected from multiple vehicles and mobile devices, these studies evaluate model suitability for real-world conditions. Several works also investigate multisensor fusion using cameras, IMUs, and

GPS to improve detection reliability and localization accuracy, reducing false positives and providing richer metadata [2], [10].

Despite methodological advances such as dilated convolutions, improved data augmentation, and optimized edge inference [5], [11], gaps persist in benchmarking and evaluation. The field lacks standardized, large-scale, cross-city datasets with consistent severity labels and geospatial ground truth, complicating cross-study comparisons [8], [9]. Furthermore, although some prototype systems demonstrate end-to-end pipelines from detection to reporting, rigorous evaluations of long-term maintenance outcomes, cost–benefit analyses, and organizational impacts remain limited [6], [10].

Building on these limitations, the proposed SpotNFix system advances pothole management beyond detection. Unlike existing approaches that focus primarily on sensor-based or image-based identification, SpotNFix integrates YOLOv11n for real-time detection with MobileNet for severity classification, enabling damage prioritization. The system further introduces a comprehensive management ecosystem, including role-based dashboards for citizens, contractors, and authorities, transparent bidding mechanisms, community funding options, and accountability tools such as fines and real-time repair tracking. By bridging the gap between detection, reporting, and repair, SpotNFix addresses critical institutional and operational shortcomings in prior systems, offering a more scalable, transparent, and effective solution for urban road infrastructure maintenance.

Year	Domain	Algorithm	Dataset	Performance Metric
[1] 2025	A Real-Time Pothole Detection System for Road Damage Detection and Classification — A. Manaswiprasad et al.	Road damage / pothole detection and classification	Custom-trained Deep Neural Network (realtime from mobile/vehicle cameras)	Custom dataset from road images (size not specified)
[8] 2024	Road Damage Detection and Reporting System Using YOLOv7 — N. Nizamiyati et al.	Road damage detection & reporting	YOLOv7 (real-time object detection)	Custom Lampung dataset (~1200 images with geotags)
[10] 2023	Deep Learning Based Pothole Detection — D. R. Rajan et al.	Pothole detection	Deep Learning (architecture not clearly specified)	Dataset details not clearly available
[2] 2021	Severity Estimation of Potholes in Imagery Using CNN — V. Kok et al.	Pothole detection & severity estimation	Faster R-CNN + CNN (VGG16, VGG19, ResNet50)	399 web images + custom Johannesburg dataset (with depth labels)
[3] 2021	A Modern Pothole Detection Technique using Deep Learning — A. Kumar et al.	Road surface defect detection	Faster R-CNN + Inception-V2	Image/video dataset (not clearly specified)
[5] 2021	Real-time Machine Learning-Based Approach for Pothole Detection — O. A. Egaji et al.	Road damage detection (sensor-based)	One-class SVM (sensor data based)	Real-world vehicle sensor data (bus deployment)
[6] 2021	Smart Pothole Detection using Dilated Convolution — K. R. Ahmed	Pothole detection	Deep Learning (Dilated Convolution)	Dataset not standardized or publicly benchmarked
[11] 2019	Asphalt Pavement Pothole Detection using YOLO — E. N. Ukhwah et al.	Pothole detection	YOLO-based deep learning (YOLOv3 variants)	Road image dataset (size not specified)
[9] 2019	Smart Pothole Detection using Vehiclemounted Sensors — A. Anaissi et al.	Pothole detection (sensor-based)	One-class SVM (VRNI system)	Vehicle sensor data (real-world roads)

[7] **2017**	PADS: A Reliable Pothole Detection System — J. Ren & D. Liu	Pothole detection (sensor-based)	K-means clustering + thresholding	Accelerometer/sensor data
[4] **2016**	Pothole Reporting System — E. Hammell et al	Pothole detection & reporting	Accelerometer-based sensing system with Android app	Sensor data from vehicles

Table 2.1 : Literature Survey of the previous studies

Chapter 3

Limitation of Existing System

Despite the availability of digital complaint platforms and mobile applications for reporting road damage, existing pothole management systems suffer from several operational and structural limitations. These limitations reduce efficiency, transparency, and accountability in infrastructure maintenance.

A. Delayed Action and Inefficient Resolution

In many existing systems, even after a pothole is reported, corrective action may take weeks or months. The absence of automated prioritization mechanisms results in delayed repairs, particularly for high-risk potholes that require immediate attention.

B. Lack of Accountability

Current platforms often mark complaints as “resolved” without proper verification of actual repair work. There is no strict monitoring framework or enforcement mechanism to ensure long-term durability of repairs, leading to recurring potholes and misuse of public funds.

C. Limited Geographic Coverage

Several pothole reporting applications operate only within specific cities or states. The absence of a scalable and nationwide digital infrastructure limits uniform adoption and prevents standardized road maintenance practices.

D. Absence of Automated Machine Learning Verification

Most existing systems lack automated AI/ML-based validation. As a result, false, duplicate or irrelevant reports may be assigned for repair, increasing operational overhead.

E. Technical and Usability Issues

Many municipal applications suffer from technical glitches such as frequent crashes, inaccurate GPS tagging, image upload failures, and lack of real-time updates. These issues discourage citizen participation and reduce trust in digital governance platforms.

F. Absence of Contractor Accountability and Penalty Mechanisms

Contractors are rarely penalized for substandard or temporary repair work. The lack of warranty-based monitoring mechanisms allows repaired potholes to reappear without consequences, leading to repeated expenditure of public funds.

G. No Transparent Bidding Framework

Most municipal systems do not include a competitive bidding mechanism for pothole repair tasks. The absence of transparent bidding reduces cost efficiency and may lead to biased contractor selection.

H. Lack of Severity-Based Prioritization

Repairs are generally not prioritized based on pothole severity or risk level. Without structured classification, minor damages may receive attention before critical road hazards, compromising public safety.

Chapter 4

Problem Statement and Objective

4.1 Problem Statement

Urban road infrastructure in rapidly growing cities like Mumbai is severely impacted by potholes caused by heavy traffic, poor construction quality, and extreme weather conditions such as monsoons. Existing pothole reporting systems rely heavily on manual complaints, which are often unverified, duplicated, and lack accurate geolocation data. These inefficiencies lead to delayed repairs, poor prioritization, and lack of accountability among authorities and contractors. Furthermore, traditional systems do not incorporate automated validation or severity assessment, making it difficult to identify critical road hazards in real time.

We propose SpotNFix, a lightweight, AI-driven, web-based system that enables real-time pothole detection using YOLO-based object detection models, followed by severity classification using a MobileNet-based Convolutional Neural Network (CNN). The system integrates geolocation services, transparent contractor bidding, workflow management, and real-time dashboards to ensure efficient, accountable, and data-driven road maintenance without relying on complex or costly infrastructure

4.2 Objectives

1. To develop an AI-driven system for automatic pothole detection using deep learning model.
2. To implement a severity classification mechanism (Minor, Medium, Major) using MobileNet CNN for effective prioritization of pothole repairs.
3. To design a web-based platform that allows citizens to easily upload pothole images with geolocation data.
4. To reduce false and duplicate reports through automated AI-based image verification.
5. To enable real-time tracking and monitoring of pothole status using dashboards and live maps.
6. To create a transparent workflow system connecting citizens, government authorities, and contractors.
7. To implement a contractor bidding mechanism for cost-effective and accountable repair allocation.
8. To improve repair prioritization by focusing on severity and risk level instead of manual complaint order.
9. To enhance accountability and transparency through post-repair verification and reopening mechanisms.
10. To transform traditional pothole management into a data-driven and proactive infrastructure system.

4.3 Scope

1. Integrated Detection and Repair Workflow

Unlike existing systems that focus only on pothole detection or reporting, the scope of SpotNFix includes a complete lifecycle approach—from AI-based verification using YOLO and severity classification using MobileNet CNN to contractor bidding, repair tracking, and accountability enforcement within a unified platform.

2. AI-Based Image Verification and Severity Classification

The system utilizes YOLO-based deep learning models for accurate pothole detection and a MobileNet-based CNN for severity classification (Minor, Medium, Major). This two-stage pipeline enables reliable verification of reports and prioritization of repairs based on the severity and risk level of potholes.

3. Performance Evaluation

• Detection performance (YOLO) will be evaluated using metrics such as Precision, Recall, F1- score, and mAP.

• Severity classification (MobileNet CNN) will be evaluated using accuracy, precision, recall, and F1-score across severity classes.

• System-level performance will be measured based on reduction in false reports, improved repair prioritization, efficiency of contractor allocation, and overall workflow transparency.

4. Real-Time Monitoring and Decision Making

The system is designed to operate in real time, continuously monitoring reported potholes, performing AI-based verification, classifying severity using MobileNet CNN, and dynamically updating repair workflows through dashboards, live maps, and status tracking systems.

Chapter 5

Proposed System

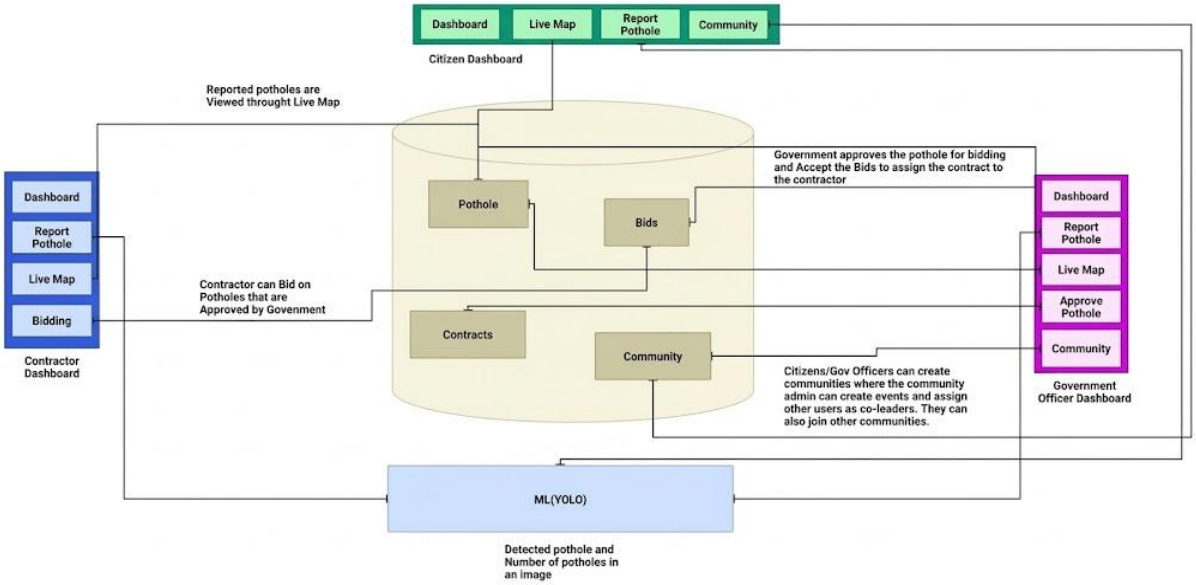

Figure 5.1 : Architecture of the Proposed System

SpotNFix is an AI-driven pothole reporting and management platform designed to enhance efficiency, transparency, and accountability in urban road maintenance. The system integrates machine learning-based pothole detection with a structured governance workflow involving citizens, government officials, contractors, and administrators.

5.1 System Overview

SpotNFix follows a role-based architecture consisting of three primary user categories:

- Citizens – Report potholes and monitor repair status
- Government Officers – Validate reports and manage contractor assignments
- Contractors – Bid for and execute repair work

Each user interacts with the system through dedicated dashboards, ensuring structured access control and workflow transparency.

5.2 Citizen Module Citizens act as primary data contributors.

They can:

- Upload pothole images through a web interface
- Automatically attach GPS-based location data
- Track pothole status (Unverified, Verified, Under Review, Repaired, Reopened, Discarded)
- View potholes on a live interactive map
- Access analytics reports showing repair progress

Initially, all submitted potholes are marked as Unverified.

5.3 Government Officer Module

Government officers are responsible for validation and administrative control. Their functions include:

- Reviewing uploaded pothole images
- Approving or discarding reports
- Initiating contractor bidding for verified potholes
- Approving the most suitable contractor (typically cost-based and contractor history based selection)
- Verifying post-repair proof images

If approved, the pothole status transitions to Verified, then to Under Review, and finally to Repaired after confirmation. If rejected, the pothole is marked as Discarded.

5.4 Contractor Module

Contractors interact with the system through:

- Viewing verified potholes open for bidding
- Submitting cost quotations

- Uploading post-repair images as proof
- Receiving approval or reopening notifications

If repair quality is unsatisfactory, the case is marked as Reopened, requiring corrective action. Additionally, if a citizen reports a pothole at a location previously marked as repaired, the system automatically reopens the case to enforce accountability.

5.5 AI-Based Verification and Automation

To reduce false reports and manual workload, SpotNFix integrates a YOLO-based machine learning model that:

- Detects potholes in uploaded images
- Counts detected instances
- Filters irrelevant or low-quality submissions

This automated verification enhances reliability and reduces administrative burden.

5.6 Transparency and Community Engagement

The system includes:

- A live map displaying pothole locations and status
- Real-time reporting dashboard
- Community module for civic engagement activities
- Event creation and participation (road safety drives, awareness programs)

This promotes collaborative urban infrastructure management.

5.7 Pothole Lifecycle

1. A pothole is first marked Unverified, then reviewed by an officer. If valid, it becomes Verified; otherwise, it is Discarded.
2. Once verified, contractors bid for the repair, and the selected bid moves the task to Under Review (repair in progress).
3. After completion, it goes through a Final Review by authorities. If approved, it is Finalized; if not, it returns to Under Review with penalties.
4. If a pothole reappears at the same location, the case is Reopened, the contractor may be penalized, and the process repeats.

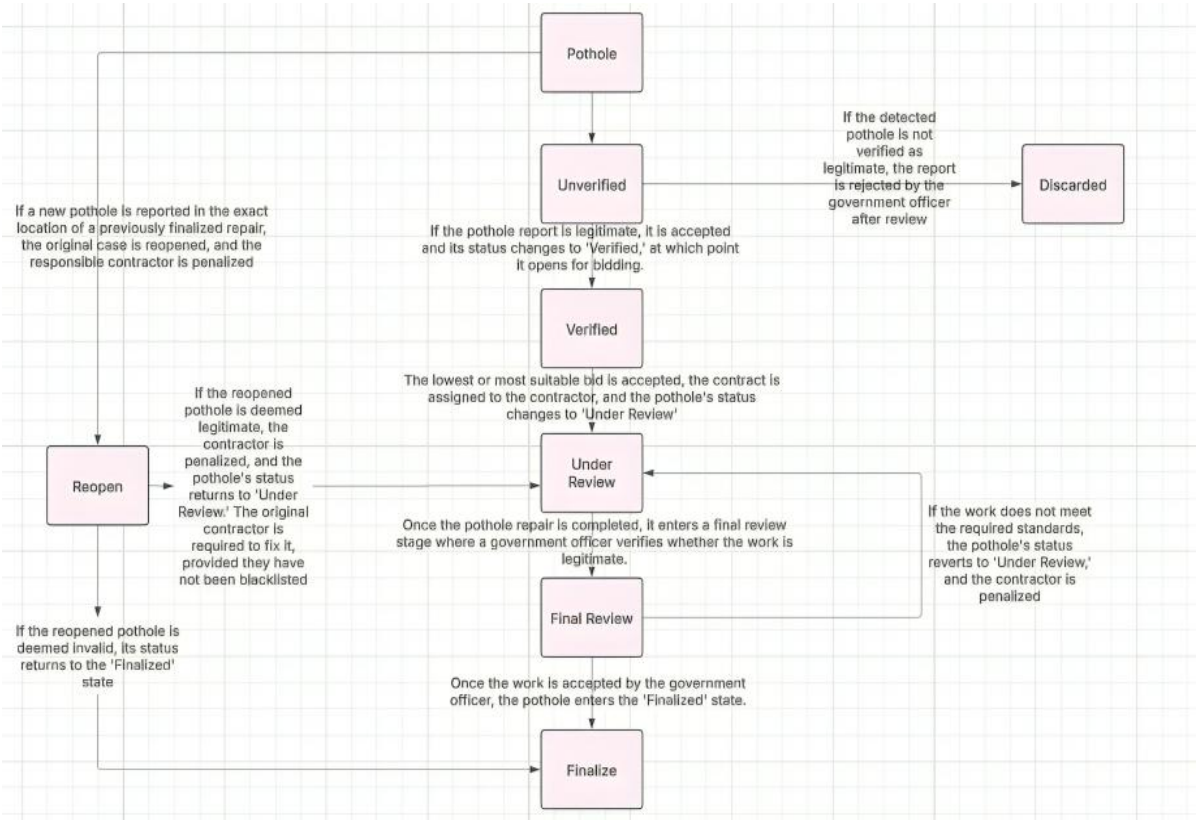

Figure 5.7.1 : Pothole Lifecycle

5.8 Dataflow Diagram:

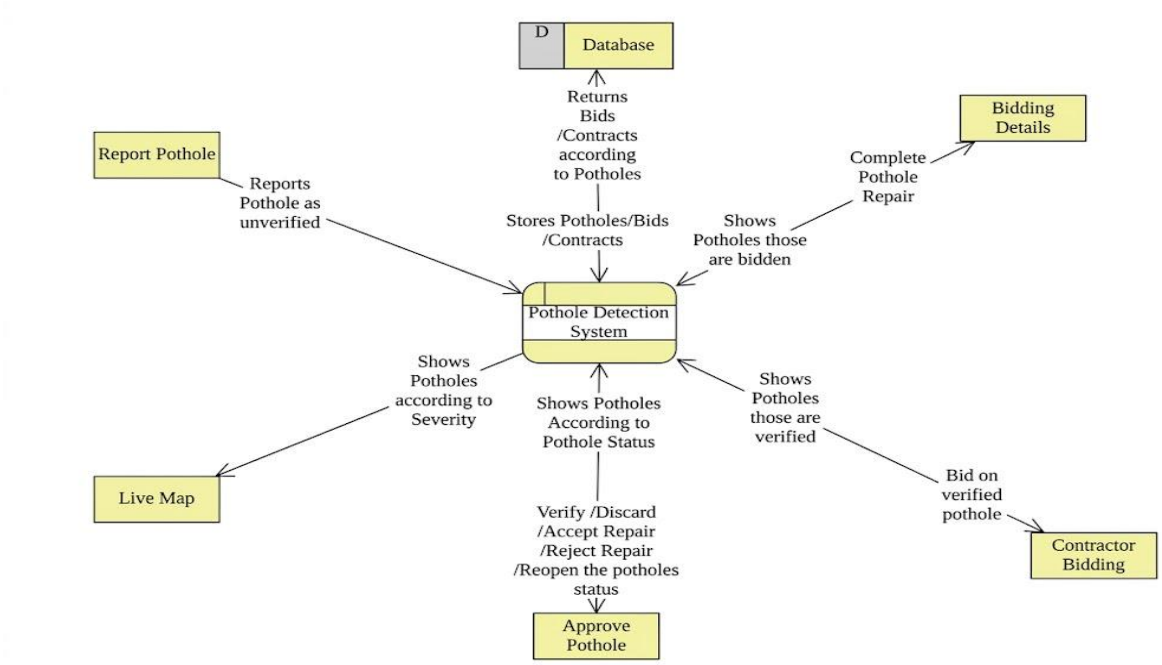

Figure 5.8.1 : Data Flow Diagram Level-0

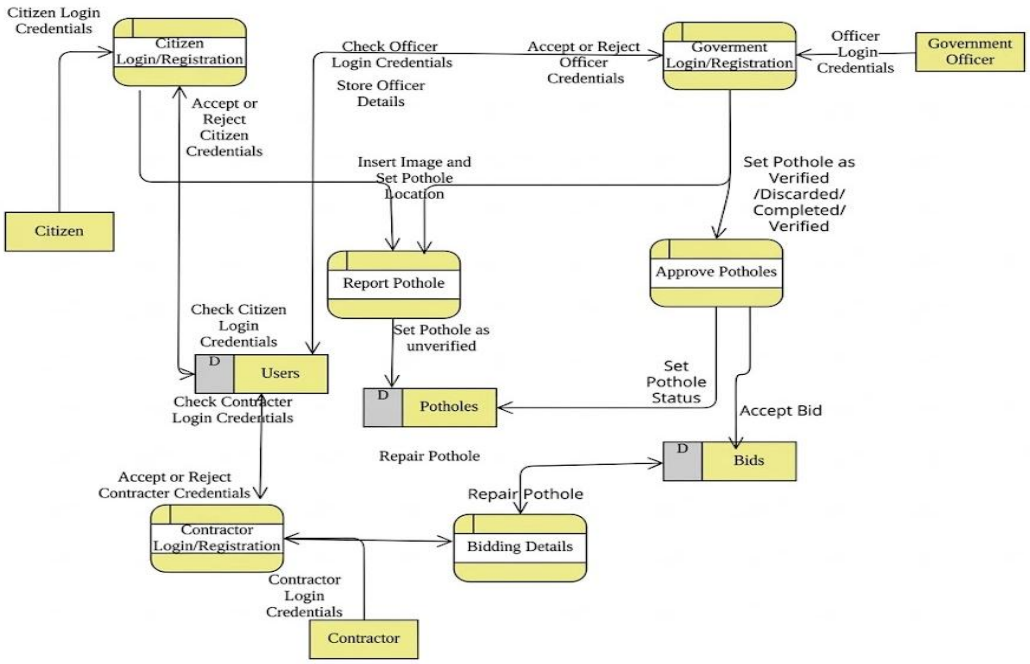

Figure 5.8.2 : Data Flow Diagram Level-1

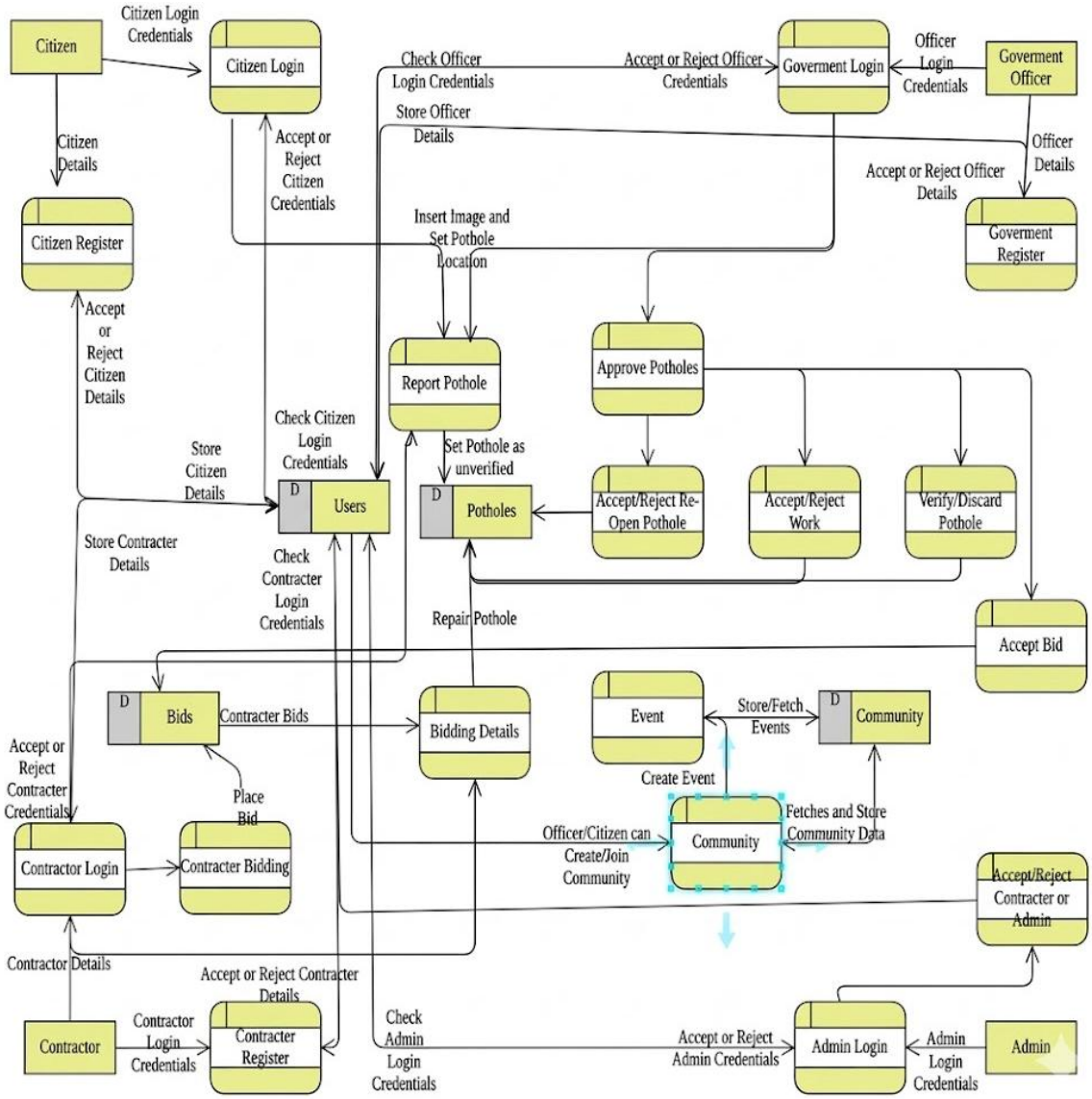

Figure 5.8.3 : Data Flow Diagram Level-2

5.9 Use Case Diagram

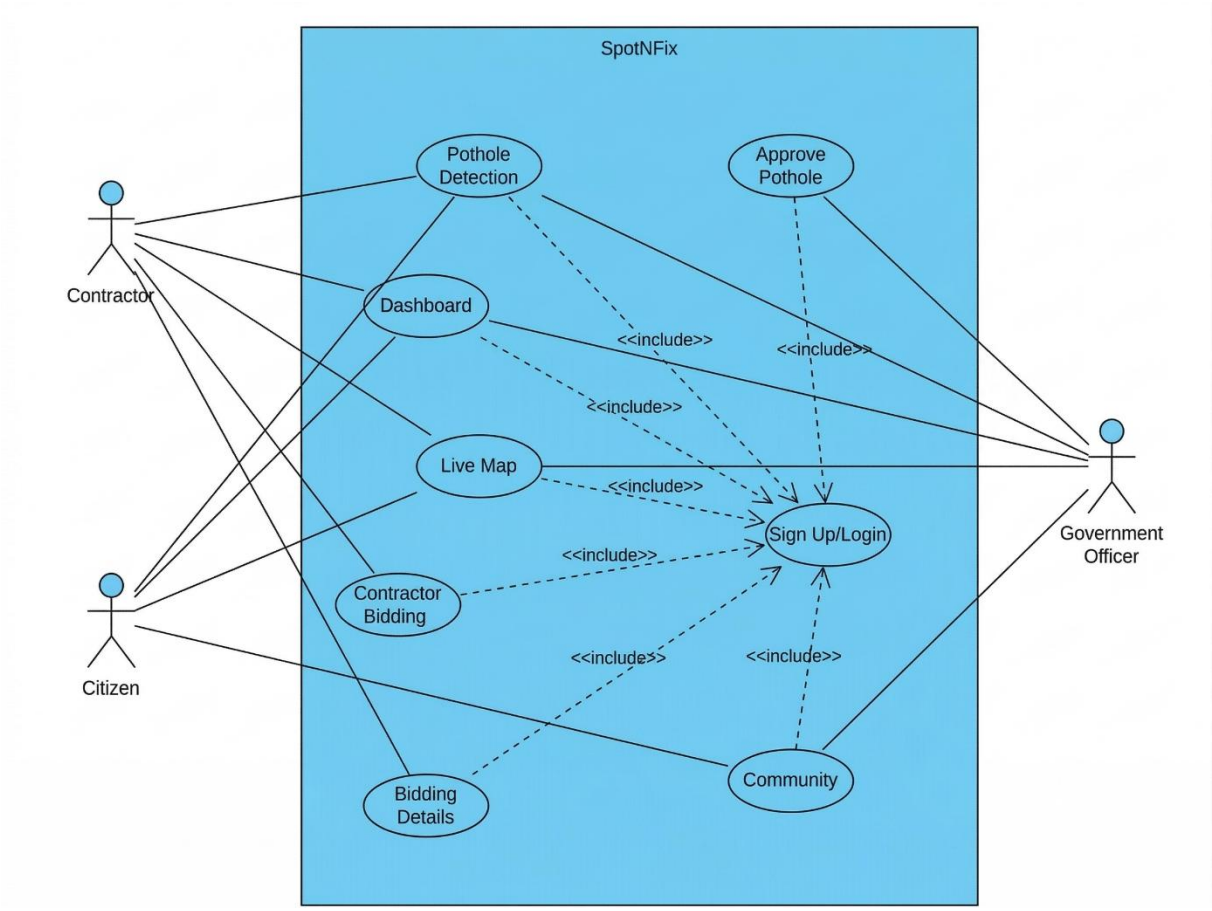

Figure 5.9.1 : Use Case Diagram

5.10 Case Study and Methodology

5.10.1 Case Context

This study focuses on urban road infrastructure management in rapidly developing Indian cities characterized by:

- High traffic density
- Monsoon-induced road degradation
- Frequent pothole complaints

Metropolitan and semi-metropolitan cities such as Mumbai, Pune, and Thane were considered representative environments due to:

- Large population density
- Mixed residential-commercial road networks
- Recurring pothole-related civic complaints

These cities experience accelerated road surface deterioration during monsoon seasons, creating a practical and relevant environment for evaluating AI-driven pothole management systems.

Despite existing municipal complaint portals, inefficiencies persist due to:

- Duplicate submissions
- Inaccurate geotagging
- Manual verification burden
- Delayed resolution

This context motivated the implementation of the SpotNFix framework.

5.10.2 Methodological Framework

The study adopts a system-oriented and application-driven methodology, combining:

1. Qualitative analysis of existing civic workflows
2. Quantitative machine learning-based detection techniques

The methodology follows a structured pipeline:

1. Image Capture and Upload
2. Automated Geolocation Association
3. AI-Based Verification
4. Severity Assessment
5. Bidding and Repair Workflow
6. Post-Repair Verification

5.10.3 YOLO-Based Pothole Detection Methodology

1) Model Architecture

The pothole detection system is trained as a single-class object detector using the Ultralytics YOLO framework. A lightweight pretrained backbone (YOLO11-nano) is selected to balance computational efficiency and detection accuracy. The model is initialized from yolo11n.pt, enabling fast inference suitable for real-world deployment.

2) Training Configuration

Model training parameters:

- Input Resolution: 640 × 640
- Epochs: 100 •Batch Size: 8
- Optimizer: AdamW
- Initial Learning Rate: 0.01
- Weight Decay: 0.0005
- Cosine Learning Rate Scheduler: Enabled
- Warmup Epochs: 3
- Early Stopping Patience: 25 epochs

GPU acceleration is utilized when available. Automatic Mixed Precision (AMP) and dataset caching are enabled for efficiency.

3) Data Augmentation Strategy

To enhance generalization and mitigate overfitting, the following augmentations are applied:

Photometric Augmentations:

- HSV Hue: 0.015
- HSV Saturation: 0.7
- HSV Value: 0.4

Geometric Augmentations:

- Rotation: ±10°
- Translation: 0.1
- Scaling: 0.5
- Shear: 2.0
- Horizontal Flip: 0.5

Mosaic augmentation is enabled during early training and disabled during the final 10 epochs to refine natural image predictions. A dropout rate of 0.1 is incorporated to improve generalization.

4) Deployment and Optimization

After training, the model is exported to ONNX format (opset 12) with dynamic input shapes enabled. For efficient CPU deployment:

- ONNX Runtime is used for inference
- Optional INT8 dynamic quantization can be applied

This enables scalable deployment in resource-constrained environments.

5.10.4 MobileNetV2-Based Severity Classification

1) Model Architecture

The classification model uses a MobileNetV2 backbone pretrained on ImageNet with an input resolution of 224 × 224. The classification head consists of Global Average Pooling followed by fully connected layers with Dropout and Batch Normalization for regularization. The final dense layer outputs three classes using softmax activation. The model was trained using categorical cross-entropy loss and the Adam optimizer.

2) Training Strategy

A two-phase training strategy was adopted. In the first phase, the MobileNetV2 backbone was frozen and only the classifier head was trained with a learning rate of 0.001. In the second phase, the entire network was fine-tuned with a reduced learning rate of 0.0001. Regularization

was applied using Dropout layers and Batch Normalization. Training stability and generalization were improved using EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint callbacks. To ensure reproducibility, fixed random seeds were used for NumPy and TensorFlow.

3) Data Handling and Augmentation

A group-aware split stratified by maximum severity was used to prevent leakage across datasets. Augmentation was applied only to the training set and included rotation, shifting, shearing, zooming, horizontal flipping, and brightness adjustments. The preprocessing pipeline followed: Bounding box crop with margin → square padding → resizing to 224 × 224 → MobileNetV2 preprocessing.

Chapter 6

Experimental Setup

The experimental evaluation is conducted to assess the effectiveness of the proposed SpotNFix system in improving pothole detection accuracy, severity classification, and overall repair workflow efficiency in urban environments. The experiments are divided into two stages corresponding to the core components of the system: AI-based pothole detection and severity classification.

Pothole detection is formulated as an object detection task using a YOLO-based deep learning model trained on annotated road image datasets. The dataset consists of labeled pothole images collected from publicly available sources and preprocessed to ensure consistency in resolution, orientation, and annotation format. The model is trained to accurately localize potholes in real-world conditions while minimizing false positives. Standard dataset splitting techniques are applied to divide the data into training, validation, and testing sets.

For severity classification, a MobileNetV2-based Convolutional Neural Network (CNN) is used to categorize detected potholes into multiple severity levels such as minor, medium, and major. The classification model is trained on cropped pothole regions extracted from the detection stage, ensuring focused learning on relevant features. Data augmentation and preprocessing techniques are applied to improve generalization and robustness across varying lighting and road conditions.

All models are trained and evaluated in a scalable computing environment, ensuring reproducibility and efficient deployment for real-time urban infrastructure monitoring.

6.1 Dataset Details

The performance of the proposed SpotNFix system is evaluated using two distinct datasets corresponding to the two-stage pipeline: pothole detection and severity classification.

6.1.1 Dataset for Pothole Detection (YOLO Model)

Training data for pothole detection were sourced from the Roboflow Universe – Pothole v1 dataset, licensed under CC BY 4.0. The dataset consists of annotated road images specifically prepared for object detection tasks.

The key specifications are as follows:

• Total Images: 2,771 annotated images

• Classes: 1 (Pothole)

• Annotation Format: YOLO format

• Image Resolution: Resized to 640 × 640 pixels

• Data Split: Training, Validation, and Testing sets

Automatic preprocessing techniques were applied to ensure data consistency, including EXIF metadata removal, image orientation correction, and standard resizing. These steps improve model robustness and ensure compatibility with the YOLO-based detection pipeline.

6.1.2 Dataset for Severity Classification (MobileNet CNN)

For severity classification, the Kaggle Annotated Potholes Dataset was utilized. This dataset contains road images with annotations in Pascal VOC XML format, where each image is associated with bounding box information.

The dataset includes three severity classes: Minor, Medium, and Major, with approximately 717 annotated images. Multiple pothole samples are extracted from each image using bounding boxes to create a focused classification dataset.

A group-aware data splitting strategy was adopted to prevent data leakage, dividing the dataset into training, validation, and testing sets in a 70:20:10 ratio, based on images. Stratification was performed using the maximum severity label per image to maintain class distribution balance. Preprocessing steps include bounding box cropping with a small margin, square padding, and resizing to 224 × 224 pixels, followed by normalization compatible with the MobileNetV2 architecture. These steps ensure efficient feature extraction and improved classification performance.

6.2 Performance Evaluation Parameters (Validation Metrics)

Since SpotNFix involves both object detection and classification, multiple evaluation metrics are used instead of relying solely on accuracy. The system is evaluated at two levels: YOLO-

based pothole detection and MobileNetV2-based severity classification, along with overall system performance.

For Pothole Detection (YOLO Model)

• Precision → Correctness of detected potholes

• Recall → Ability to detect actual potholes

• F1-Score → Balance between precision and recall

• mAP@0.5 → Overall detection performance

The model achieved mAP@0.5 = 0.773, F1-score ≈ 0.75 at confidence 0.35, and high recall (0.96), indicating reliable detection performance.

Threshold Analysis

• Threshold = 0.35 (Selected) → Balanced precision and recall

• Higher thresholds → Higher precision, lower recall

For Severity Classification (MobileNet CNN)

• Accuracy, Precision, Recall, F1-Score → Classification performance

• ROC-AUC → Class separability

The MobileNetV2 model achieved 83.54% accuracy, F1 ≈ 0.835, and AUC ≈ 0.867, showing effective severity classification.

System-Level Evaluation

• Reduction in false/duplicate reports

• Improved severity-based prioritization

• Faster verification and repair workflow

• Enhanced transparency and accountability

6.3 Software and Hardware Setup

The implementation of the proposed SpotNFix system requires an integrated software stack and suitable hardware infrastructure to support real-time pothole detection, severity classification, and workflow management. The setup is designed to ensure scalability, efficient computation, and smooth user interaction across different platforms.

6.3.1 Software Setup

The system is developed using modern web and machine learning technologies to enable a seamless end-to-end pipeline:

• **Frontend:** The user interface is built using React (with Vite) and Tailwind CSS, enabling fast rendering, modular design, and responsive layouts for different devices.

• **Backend:** The backend is implemented using Node.js with Express, handling API requests, authentication, business logic, and communication between system components.

• **Database and Storage:** Supabase (PostgreSQL) is used for structured data storage, real-time updates, and file handling, ensuring efficient management of pothole reports, images, and user data.

• **Machine Learning Models**:

- YOLO (Ultralytics, ONNX format) is used for real-time pothole detection
- MobileNetV2 CNN is used for severity classification (Minor, Medium, Major)

• AI Integration: The system incorporates Google Gemini API to enhance classification and contextual analysis capabilities.

• APIs and Services:

- Google OAuth for secure user authentication
- Mapbox / Leaflet for geolocation tracking and live map visualization
- Supabase Realtime for live status updates
- Nodemailer for email notifications and OTP verification This software stack ensures efficient communication between modules, enabling real-time processing and scalable deployment.

6.3.2 Hardware Setup

The hardware requirements are designed to support both development and deployment environments while ensuring efficient execution of AI models:

• **Client Devices:**

Smartphones, tablets, or desktops equipped with a camera, GPS, and internet connectivity are used by citizens for uploading pothole images and accessing system features.

• **Server Requirements**:

- Minimum Configuration: 4-core CPU, 8 GB RAM
- Optional GPU Support: Recommended for faster YOLO-based inference and model execution
- Storage: Minimum 10 GB for datasets, models, and application data

• **Deployment Environment:**

The system can be deployed on cloud platforms or on-premise Linux servers, ensuring flexibility, scalability, and reliability for real-world usage.

The combined software and hardware setup enables SpotNFix to operate as a lightweight, scalable, and real-time AI-driven system, capable of handling large volumes of user-generated data while maintaining high detection accuracy and efficient workflow execution.

6.4 Implementation

The implementation of the proposed SpotNFix system demonstrates measurable improvements in pothole detection efficiency, workflow transparency, and repair prioritization. The evaluation focuses on both system-level performance and machine learning model effectiveness.

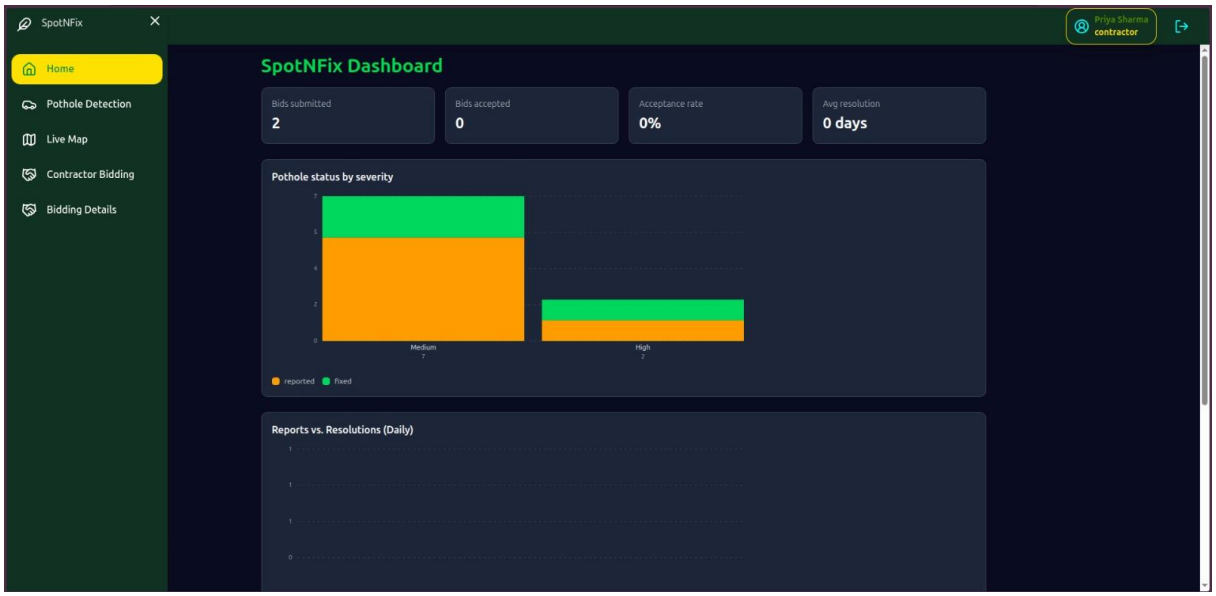

Figure 6.4.1 : Contractor Dashboard Interface

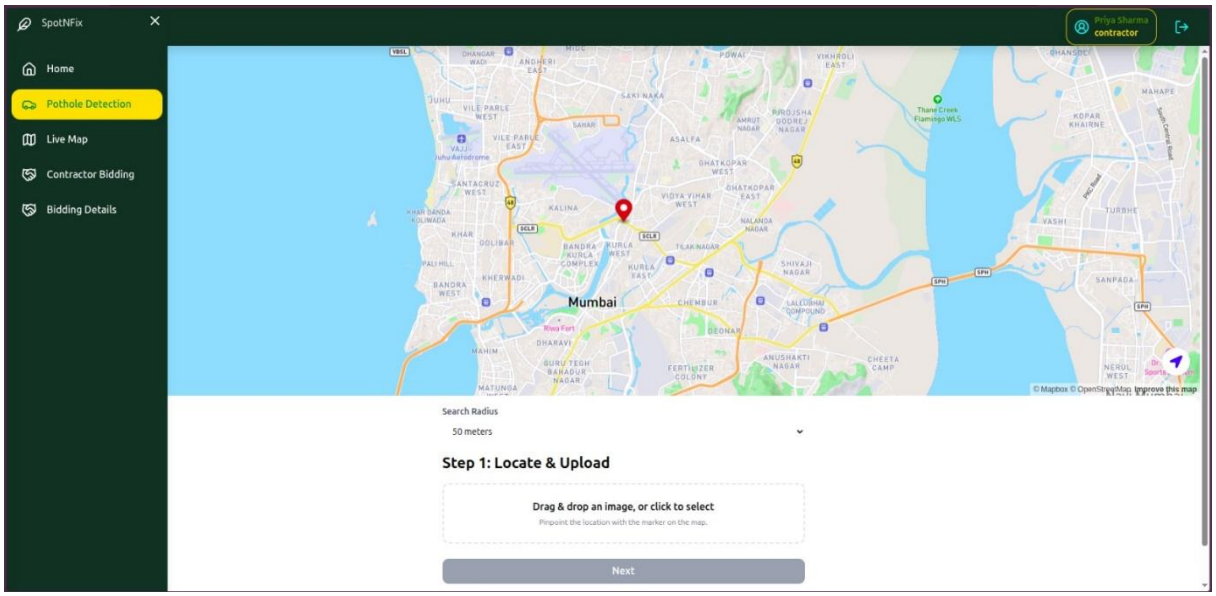

Figure 6.4.2 : Pothole Detection & Upload Interface

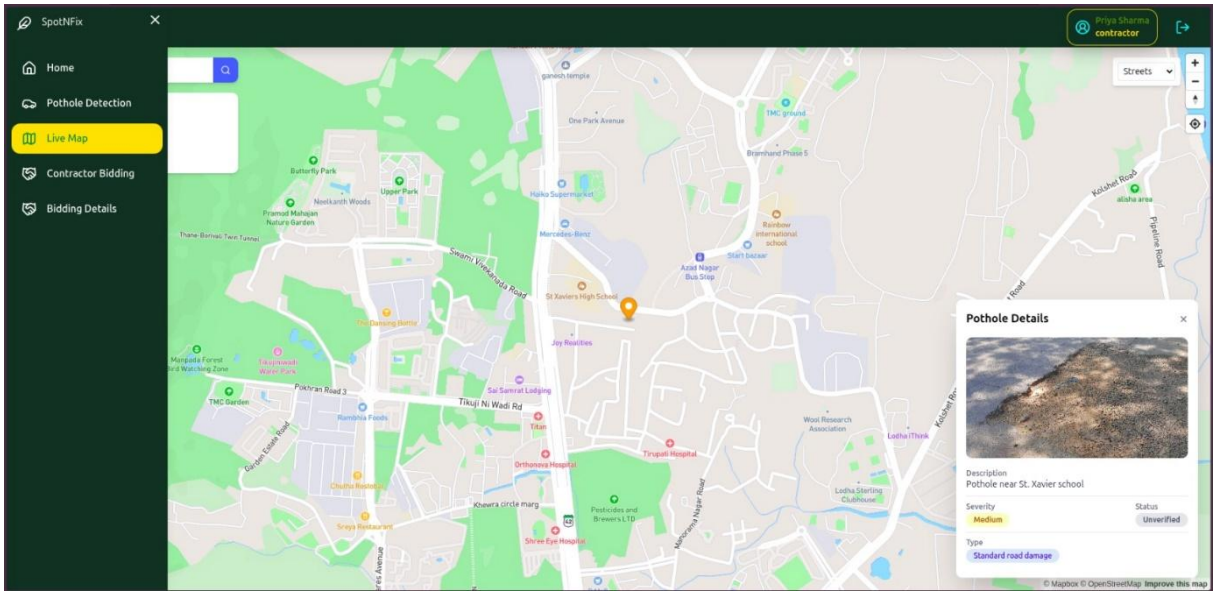

Figure 6.4.3 : Live Map View with Pothole Details

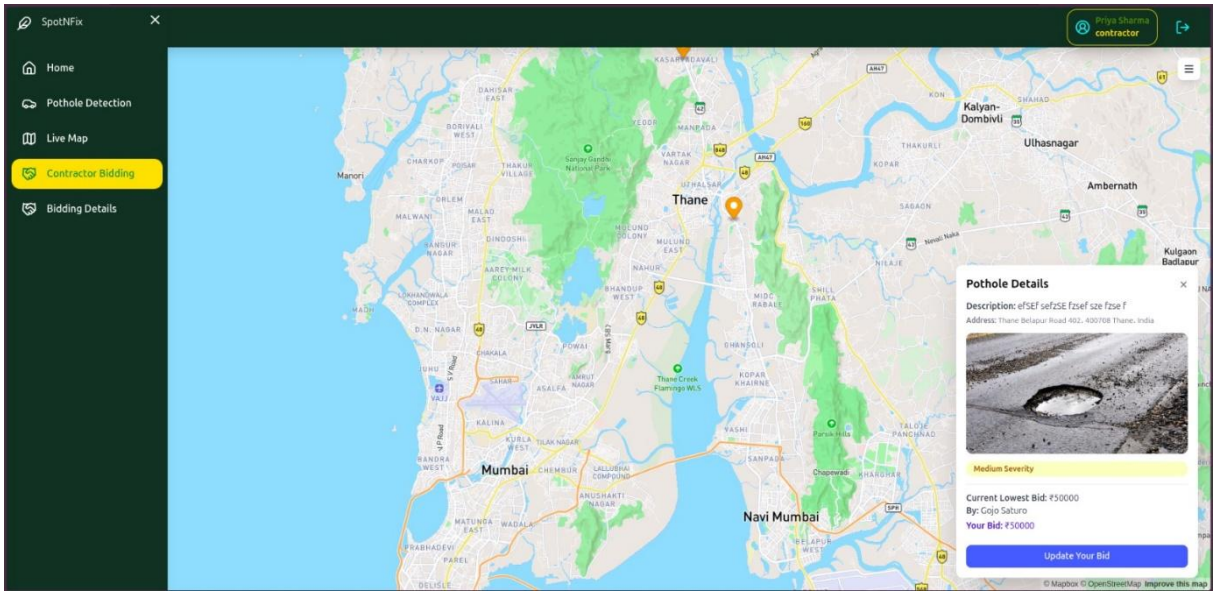

Figure 6.4.4 : Contractor Bidding Interface

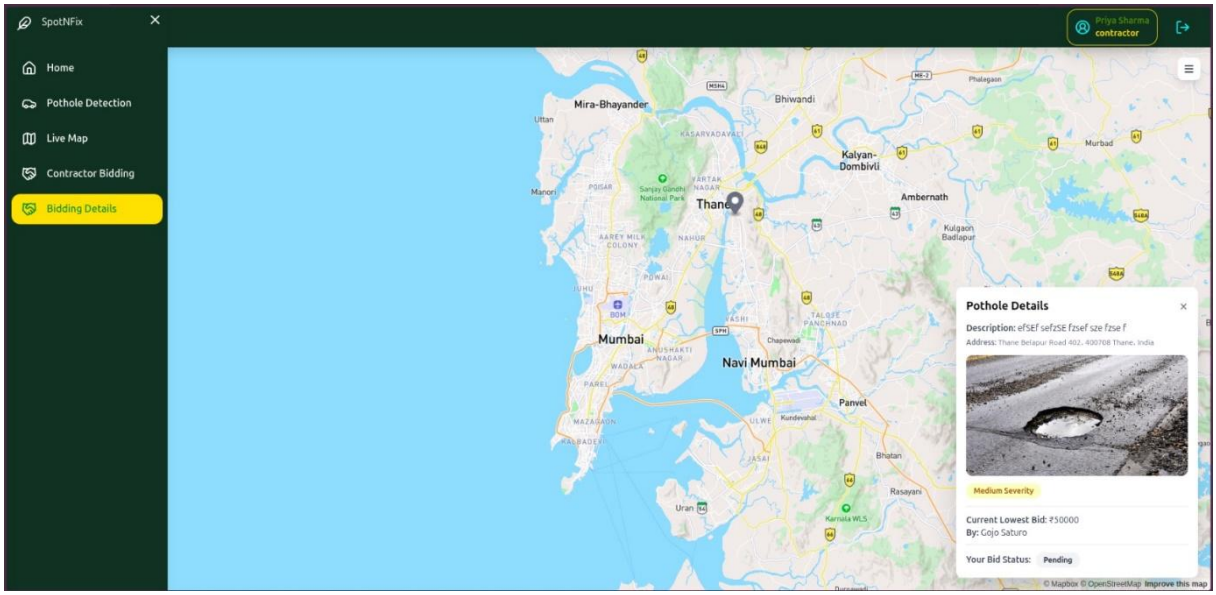

Figure 6.4.5 : Bidding Details Interface

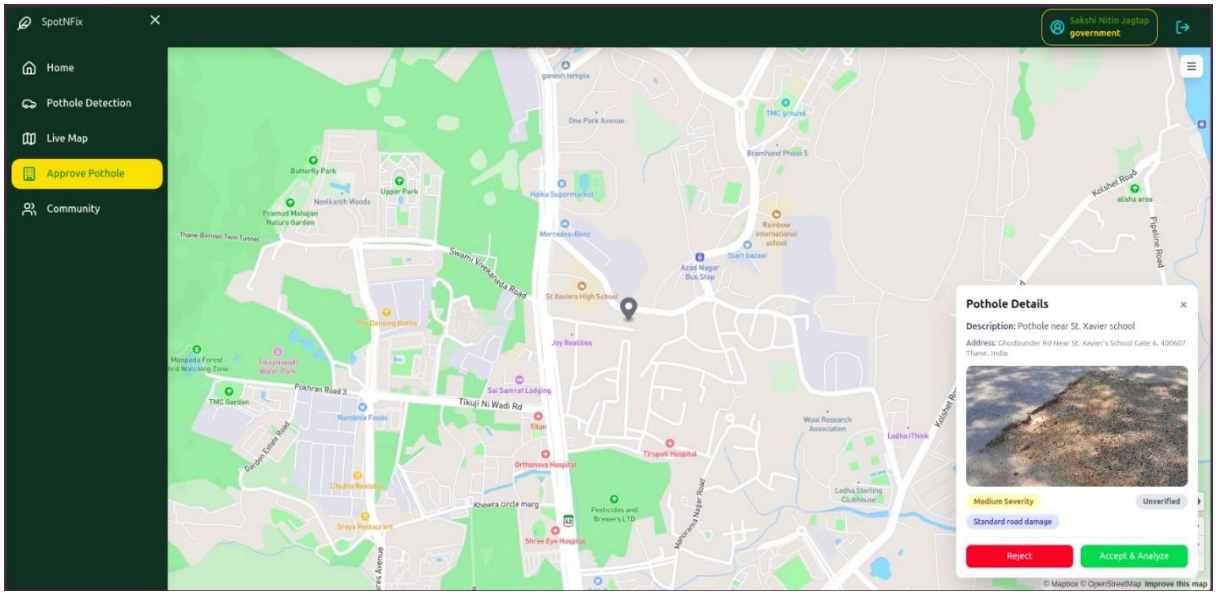

Figure 6.4.6 : Approve Pothole Interface

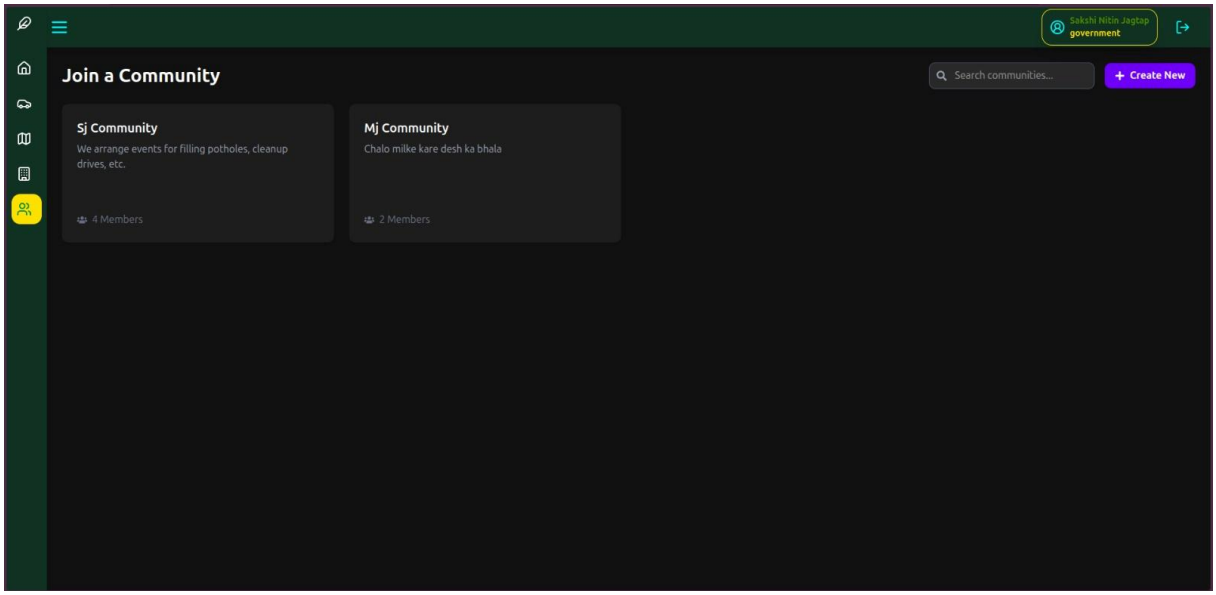

Figure 6.4.7 : Community Discovery Interface

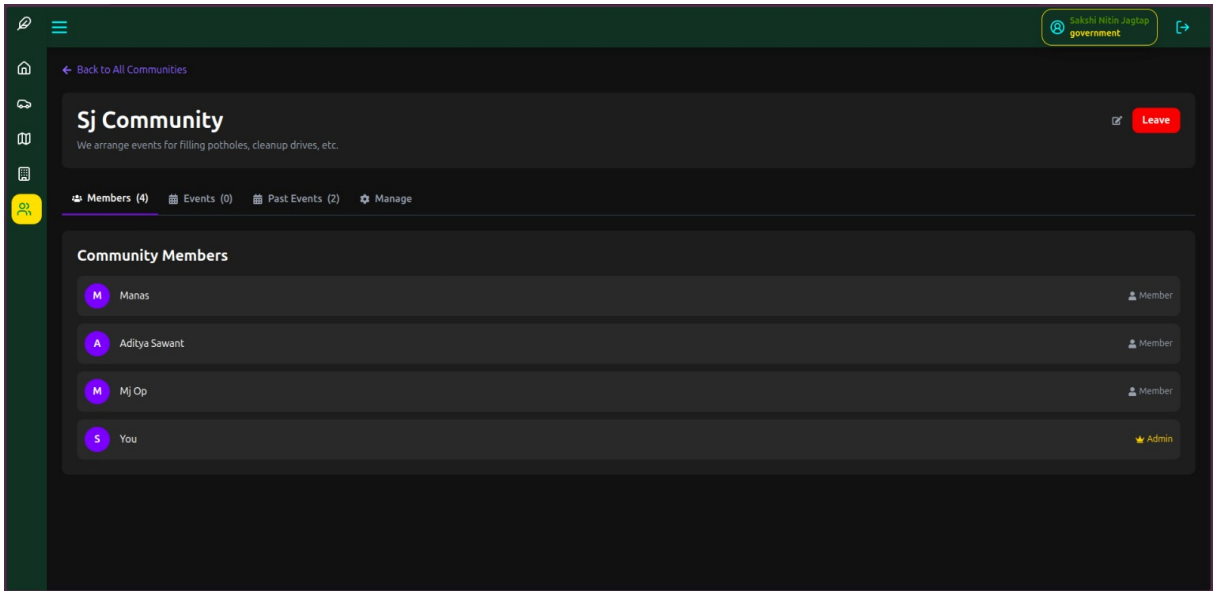

Figure 6.4.8 : Community Management Interface

Chapter 7

Results and Discussion

7.1 System-Level Improvements

The integration of AI-based image verification significantly reduced false and duplicate submissions, a common limitation in conventional complaint-driven systems. Automated filtering ensured that only validated pothole reports were forwarded for administrative approval, thereby reducing manual workload and processing delays.

Severity-based classification enabled prioritized repair allocation. High-severity potholes, particularly those located on high-traffic roads and intersections, were consistently flagged for urgent action. This risk-driven allocation model improved response efficiency compared to complaint-frequency-based prioritization.

The transparent bidding framework improved contractor accountability and cost optimization. Post-repair image verification strengthened quality assurance mechanisms, while real-time dashboards enhanced visibility into repair lifecycle stages.

Overall, the system transitions pothole management from a reactive manual process to a structured, data-driven governance model.

7.1.1 Reduction in Manual Effort through System Automation

Activity	Traditional Method (Manual)	SpotNFix System (Automated)
Pothole Identification	Field inspection required	Image upload by user
Data Recording	Manual entry by Govt. official	Auto-generated from input
Location Capture	Manual noting / estimation	GPS-based auto capture
Severity Assessment	Human judgment	System-assisted classification
Report Submission	Physical / phone complaint	Instant online submission
Duplicate Verification	Manual checking	Automated verification
Effort Required	High	Low

Table 7.1.1 : Comparison of Manual vs Automated Pothole Management Processes

The comparison presented in Table 7.1.1 highlights the significant reduction in manual effort achieved through implementation of the SpotNFix system. In the traditional approach, pothole detection, reporting, and documentation rely heavily on manual processes such as field inspections, physical complaints, and human judgment for severity assessment. These steps are time-consuming and prone to inconsistencies.

In contrast, the proposed system automates multiple stages of the workflow. Users can directly upload images, while the system automatically captures location data, generates reports, and assists in classification. This reduces the need for continuous human intervention and minimized the chances of errors.

Overall, the automation of key processes leads to improved efficiency, faster reporting, and better scalability, making the system more suitable for handling large volumes of pothole data in urban environments.

7.1.2. Analysis of Geolocation Accuracy and Positional Variations

Pothole ID	Actual Latitude	Actual Longitude	Reported Latitude	Reported Longitude	Error Distance (meters)
P1	19.0760	72.8777	19.0761	72.8778	3 m
P2	19.0785	72.8801	19.0786	72.8802	5 m
P3	19.0742	72.8753	19.0742	72.8754	2 m
P4	19.0798	72.8820	19.0799	72.8821	6 m
P5	19.0735	72.8740	19.0736	72.8741	7 m

Table 7.1.2 : Geolocation Accuracy Analysis of Reported Pothole Coordinates

The results presented in Table 7.1.2 indicate that the positional variation between the pothole location and the reported coordinates lies within a narrow range of 2 to 8 meters, demonstrating high geolocation accuracy. The system utilizes the Mapbox API for capturing GPS-based location data, while also extracting location-related metadata from the uploaded image. By comparing these two sources, minor variations can be observed and analyzed. Since this approach is directly implemented within the system, the observed differences are primarily due to inherent precision limits in coordinate representation and synchronization between the two data sources. Overall, the results confirm that the system provides reliable and consistent location information suitable for practical pothole identification and management.

7.1.3. Comparative Cost Analysis – Traditional Estimation vs Bidding Mechanism

Pothole ID	Severity	Traditional Estimated Cost (₹)	Lowest Bid Selected (₹)	Cost Difference (₹)	Savings (%)
P1	Minor	1000	820	180	18%
P2	Medium	1500	1200	300	20%
P3	Major	2500	2050	450	18%
P4	Medium	1600	1300	300	18.75%
P5	Major	2700	2200	500	18.5%

Table 7.1.3 : Comparative Cost Analysis of Traditional Estimation and Bidding Mechanism

The comparative analysis presented in Table 7.1.3 demonstrates the effectiveness of the bidding mechanism in reducing repair costs when compared to traditional fixed-cost estimation methods. In the conventional approach, repair costs are predefined and often lack flexibility, which may lead to inefficient allocation of resources.

In contrast, the simulated bidding system allows multiple contractors to submit competitive bids for each pothole repair task. Instead of selecting purely the lowest bid, the system evaluates bids based on a combination of factors including cost, contractor history, and quality of past work. As observed, this approach consistently results in cost savings ranging between approximately 18% to 20% across different severity levels.

The simulation used realistically generated data based on typical cost ranges, contractor behaviour, observed bidding behaviour, and pothole severity to approximate real-world conditions.

Since this bidding mechanism is implemented within the system, the results highlight its potential to enhance financial efficiency and promote transparency in contractor selection. By balancing cost with performance factors, the system ensures that repair work is allocated in a cost-effective and quality-assured manner without compromising operational feasibility.

7.1.4. Reduction in time through System Automation across pothole lifecycle

Stage	**Traditional Method (Time)**	**SpotNFix System (Time)**	**Time Reduction**
Pothole Identification	1–2 days	Instant (minutes)	~90%+
Report Submission	1 day	Instant	~100%
Verification & Inspection	2–3 days	Automated (minutes)	~95%
Contractor Assignment	2–4 days	10–12 hours	~75–85%
Repair Completion Initiation	3–5 days	1–2 days	~60–70%
Total Lifecycle Completion	9–15 days	2–4 days	~70–80%

Table 7.1.4 : Time Reduction Across Pothole Management Lifecycle Using Automation

The results presented in Table 7.1.4 highlight a significant reduction in the overall time required across different stages of the pothole management lifecycle. In the traditional approach,

multiple stages such as identification, reporting, verification, and contractor assignment involve manual processes, leading to delays ranging from several days to weeks.

With the implementation of the SpotNFix system, these stages are streamlined through automation and digital workflows. Pothole reporting and identification are performed instantly through image upload, while verification and processing are completed within minutes. Additionally, the contractor assignment process is accelerated through the integrated bidding mechanism, reducing delays associated with manual allocation.

The cumulative effect of these improvements results in a substantial reduction in total lifecycle completion time, bringing it down from several days to just a few days. This demonstrates the system's ability to enhance operational efficiency and enable faster response to road maintenance issues.

7.2 Model Metrics (YOLO)

1) F1–Confidence Curve

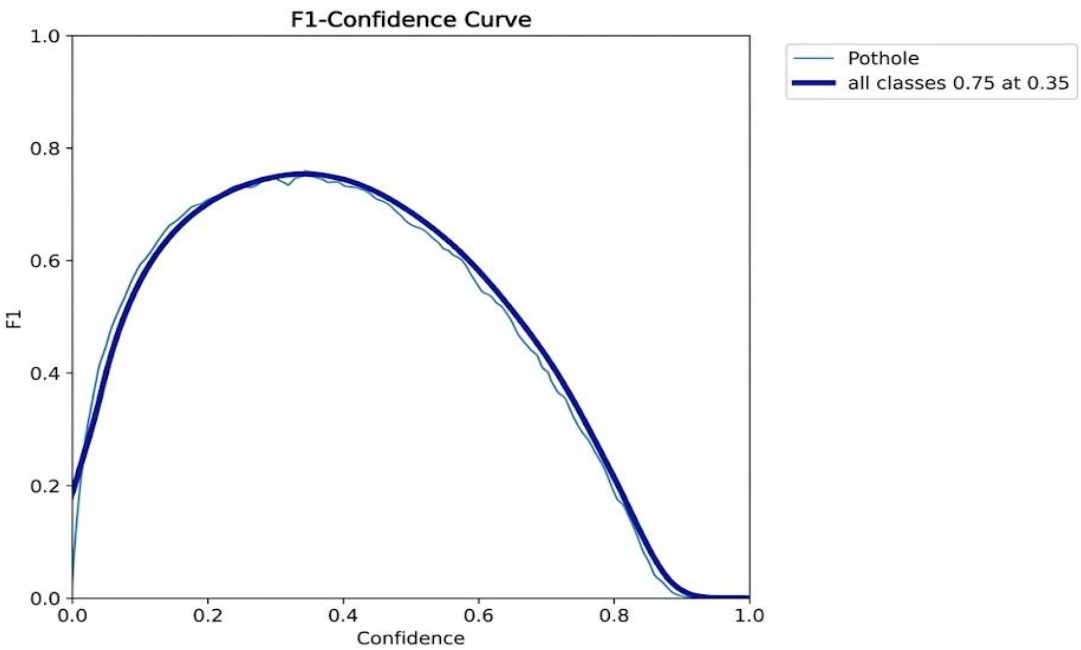

Figure 7.2.1 : F1-Confidence Curve showing optimal threshold around 0.35

The F1–Confidence curve illustrates the trade-off between precision and recall across varying confidence thresholds. The model achieves a peak F1-score of approximately 0.75 at a confidence threshold near 0.35, representing the optimal operating point for balanced detection performance.

2) Precision–Confidence Curve

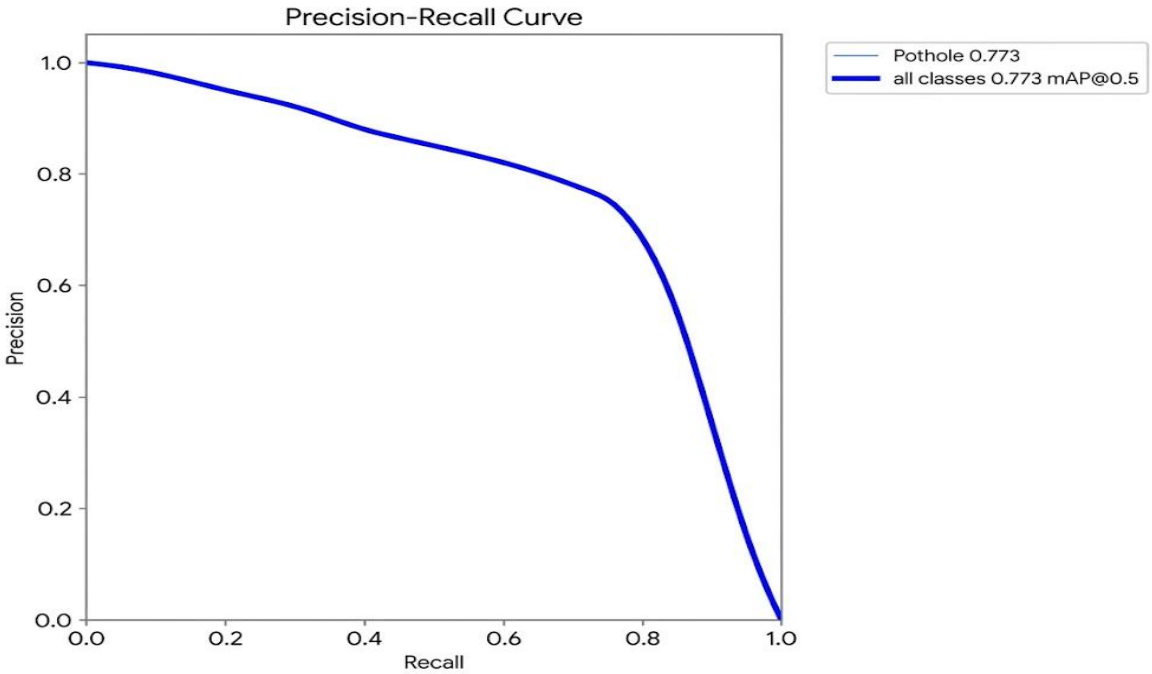

Figure 7.2.2 : Precision–Confidence Curve demonstrating increasing precision at higher threshold

The Precision–Confidence curve demonstrates a positive correlation between prediction confidence and precision. Precision steadily increases as the confidence threshold rises, reaching near-perfect precision at high confidence values. However, excessively high thresholds may reduce detection coverage.

3) Precision–Recall Curve

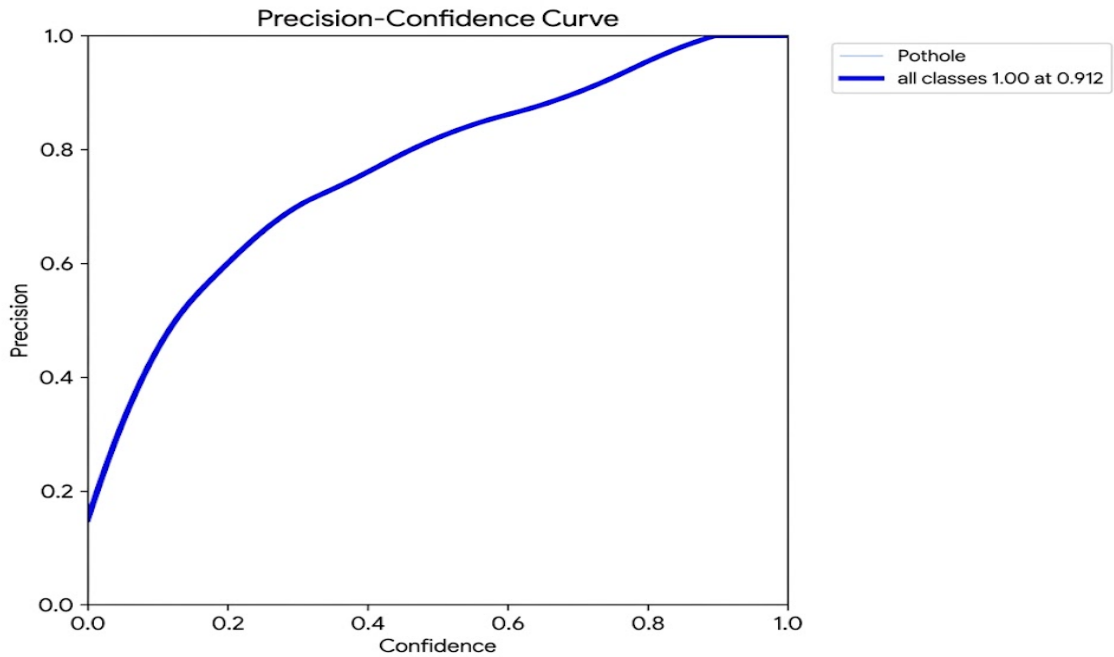

Figure 7.2.3 : Precision–Recall Curve showing mAP@0.5 of 0.773.

The Precision–Recall curve reflects the detection performance across varying recall levels. The model achieves a mean Average Precision (mAP@0.5) of approximately 0.773. High precision is maintained at moderate recall levels, confirming reliable pothole identification under balanced threshold settings.

4) Confusion Matrix

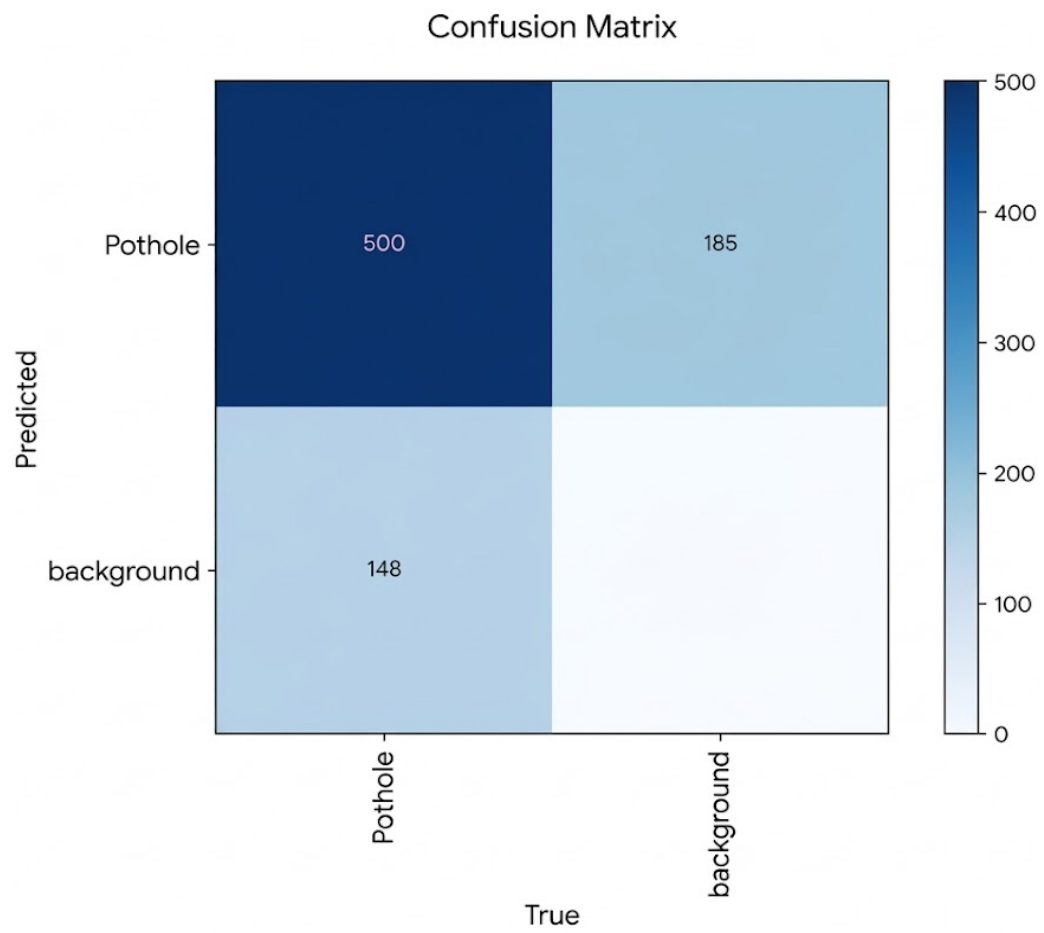

Figure 7.2.4 : Confusion Matrix showing true positives, false positives, and false negatives.

The confusion matrix summarizes classification performance between pothole and background classes. The model correctly detected a majority of pothole instances while maintaining reasonable control over false positives and false negatives. Misclassifications primarily occur under challenging lighting and shadow conditions.

7.3 Evaluation Metrics (MobileNet)

Model evaluation was performed using a confusion matrix derived from 164 test samples. The class distribution included 39 Minor, 66 Medium, and 59 Major samples. The overall accuracy achieved was 83.54%. Per-class metrics are summarized below.

Minor class achieved precision of 0.966, recall of 0.718, and F1-score of 0.824. Medium class achieved precision of 0.719, recall of 0.970, and F1-score of 0.826. Major class achieved precision of 0.978, recall of 0.763, and F1-score of 0.857.

Class	**Precision**	**Recall**	**F1-score**
Minor	**0.966**	**0.718**	**0.824**
Medium	**0.719**	**0.970**	**0.826**
Major	**0.978**	**0.763**	**0.857**

Table 7.3.1 : MobileNetV2 Severity Classification Metrics

Macro-averaged precision, recall, and F1-score were 0.888, 0.817, and 0.835 respectively. Micro-averaged metrics matched the overall accuracy due to the exclusive multi-class setting.

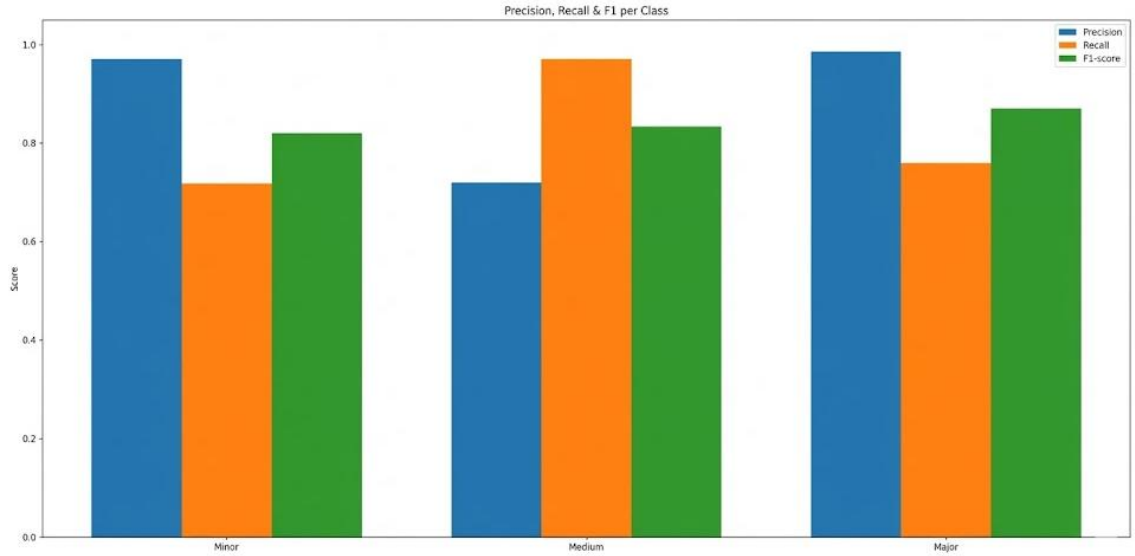

Figure 7.3.1 : Precision, Recall and F1 per class for MobileNetV2 Classification

1) ROC and AUC Analysis

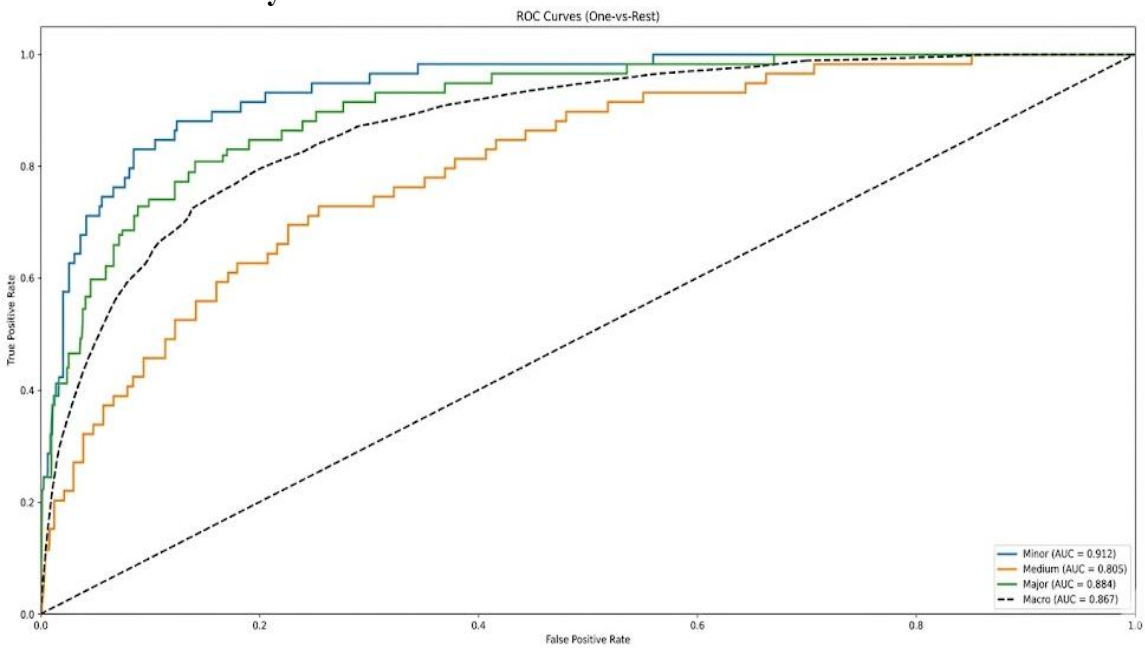

Figure 7.3.2 : ROC curves and AUC scores for MobileNetV2 Classification

Receiver Operating Characteristic (ROC) curves were computed using a one-vs-rest strategy. The AUC values obtained were 0.912 for Minor, 0.805 for Medium, and 0.884 for Major, resulting in a macro-average AUC of approximately 0.867.

Higher AUC values for Minor and Major indicate better class separability, while the Medium class remains comparatively harder to distinguish. All ROC curves lie well above the random baseline, demonstrating good discriminative performance.

2) Confusion Matrix Interpretation

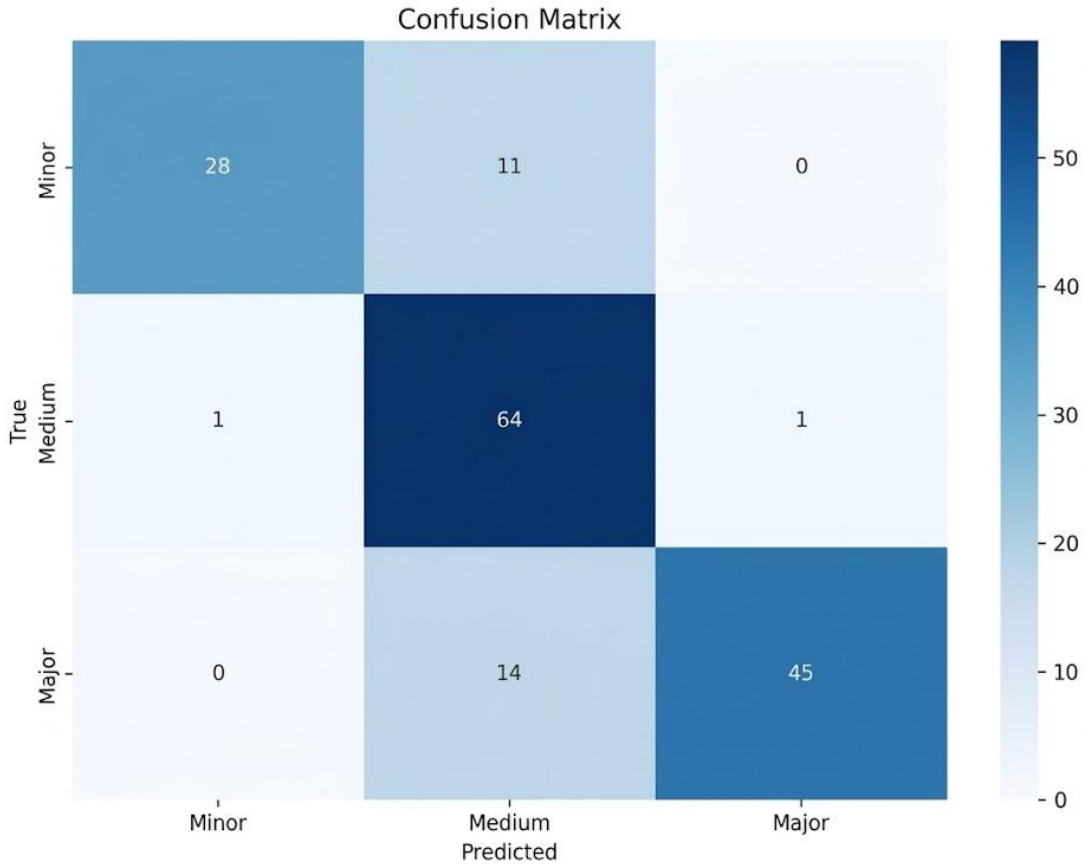

Figure 7.3.3 : Confusion matrix for MobileNetV2 severity classification.

The confusion matrix highlights that most Medium severity potholes were correctly identified (64 out of 66 samples). However, some Minor and Major potholes were misclassified as Medium. Specifically, 11 Minor samples and 14 Major samples were predicted as Medium, indicating a bias toward mid-severity predictions.

This behavior reflects a precision–recall trade-off. Minor and Major predictions exhibit high precision with lower recall, while the Medium class demonstrates high recall with comparatively lower precision. This trade-off can be tuned using threshold adjustments or class weighting depending on operational requirements.

Chapter 8

Conclusion and Future Work

The proposed system, SpotNFix, presents a comprehensive and intelligent approach to modern pothole management by integrating artificial intelligence, machine learning, and geolocation technologies into a unified digital platform. The system effectively addresses the key limitations of traditional pothole reporting mechanisms, which are often manual, inefficient, and lack verification, prioritization, and accountability. By incorporating a YOLO-based object detection model for automated pothole verification and a MobileNetV2-based convolutional neural network for severity classification, SpotNFix enables accurate detection and risk-based prioritization of road damage.

The system goes beyond detection by introducing a structured and transparent workflow that connects citizens, government authorities, and contractors. Features such as real-time dashboards, GPS-based mapping, and a bidding-based contractor allocation mechanism enhance transparency, reduce delays, and improve resource utilization. The integration of AI-based validation significantly minimizes false and duplicate reports, while severity-based classification ensures that critical potholes are addressed promptly. As observed in the system evaluation, this approach transforms pothole management from a reactive complaint-driven process into a proactive, data-driven governance model.

Furthermore, the project *SpotNFix: Smart Pothole Reporting & Severity Detection System* successfully achieved its defined objectives, demonstrating strong performance in real-world

conditions through reliable pothole detection and effective severity classification. The integration of geolocation, real-time dashboards, and AI-based verification significantly reduced false reports and improved operational efficiency. By establishing a transparent workflow connecting citizens, authorities, and contractors, along with a cost-efficient bidding mechanism, the system enhanced accountability and resource utilization. Features such as repair tracking, post-verification, and case reopening further ensured system reliability while reducing repeated repairs and unnecessary public expenditure. Overall, the system fulfills all key objectives, including detection accuracy, prioritization, workflow automation, and transparency, and aligns with smart city goals by promoting safer roads, improved civic participation, and efficient infrastructure management, despite minor limitations in challenging conditions.

Despite its effectiveness, certain limitations remain that open avenues for future enhancements. The detection performance may be affected under challenging conditions such as low lighting, shadows, or water-filled potholes, indicating the need for more diverse and large-scale datasets. Future work can focus on improving model robustness using advanced architectures, multimodal data (e.g., combining image and sensor data), and domain-specific data augmentation techniques.

Additionally, integrating real-time video-based detection from vehicle-mounted cameras or CCTV systems can further enhance coverage and reduce reliance on manual uploads. The system can also be extended to include predictive analytics, enabling authorities to identify high-risk zones and prevent pothole formation through proactive maintenance strategies. Incorporating edge deployment and mobile optimization can improve scalability and reduce latency in real-time applications.

In the long term, SpotNFix can be expanded beyond pothole detection to support broader smart city infrastructure management, including crack detection, waste monitoring, and streetlight maintenance. By continuously evolving with advancements in AI and urban analytics, the system has the potential to serve as a scalable and sustainable solution for intelligent infrastructure governance

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Appendix

A. Setup Instructions

A.1 System Requirements

Ensure the following software is installed on your system:

- Node.JS: ≥ 18.0.0 (Recommended: 20 LTS)
- Python: Version 3.10+
- Git
- Package Manager: npm or yarn
- Database: Supabase (cloud-based PostgreSQL)
- Browser: Latest version of Chrome/Edge

A.2 Cloning Repository

Step 1: Clone the Repository

$ git clone https://github.com/manastechnical/SpotNFix.git

Step 2: Navigate to Project Directory

$ cd SpotNFix

A.3 Backend Setup (Node.js + Express)

Step 1: Navigate to Backend Folder

$ cd backend

Step 2: Install Dependencies

$ npm install

Step 3: Setup Environment Variables

Create a `.env` file in the backend folder and add:

PORT=5000

SUPABASE_URL=your_supabase_url

SUPABASE_KEY=your_supabase_anon_key

JWT_SECRET=your_secret_key

Step 4: Run Backend Server

$ npm run dev

The backend server will start locally (typically at http://localhost:5000)

A.4 Frontend Setup (React + Vite)

Step 1: Open a New Terminal and Navigate to Frontend Folder

$ cd frontend

Step 2: Install Dependencies

$ npm install

Step 3: Run Development Server

$ npm run dev

The frontend will run at http://localhost:5173

A.5 ML Model Setup (Flask API)

Step 1: Navigate to ML Folder

$ cd ml

Step 2: Create Virtual Environment

$ python -m venv venv

Step 3: Activate Environment

- Windows:

  $ venv\Scripts\activate

- Linux/Mac:

  $ source venv/bin/activate

Step 4: Install Dependencies

$ pip install -r requirements.txt

Step 5: Run ML Server

$ python app.py

The ML server will run locally (typically at http://127.0.0.1:5001)

A.6 Notes

- Ensure backend, frontend, and ML server are running simultaneously.
- Configure correct API endpoints in frontend (`.env` if used).
- Supabase project must be properly set up with required tables and authentication.
- Activate the Python/virtual environment before running the ML server.
- Modify ports if conflicts occur.
- Ensure stable internet connection for Supabase services.

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SpotNFix: Smart Pothole Reporting and Severity Detection System

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Abstract: Road infrastructure maintenance is a persistent challenge in rapidly urbanizing regions, where potholes significantly contribute to road accidents, vehicle damage, and traffic inefficiencies. Conventional pothole reporting mechanisms rely on manual complaints and lack automated validation, severity assessment, and transparency in repair workflows. This paper presents SpotNFix, an AI-driven, web-based pothole reporting and management system designed to address these limitations. The proposed system enables citizens to upload geotagged images of potholes, which are automatically verified using a deep learning-based object detection model. Severity estimation is performed to prioritize repairs, while a transparent bidding mechanism ensures accountable contractor selection. By integrating machine learning, geolocation services, and role-based dashboards, SpotNFix offers a scalable, transparent, and efficient solution for smarter urban road maintenance.

Keywords: Pothole Detection, Smart Cities, YOLO, Infrastructure Management, Machine Learning

1. Introduction

Well-maintained road networks are essential for economic development, public safety, and efficient transportation. Despite continuous infrastructure investments, potholes remain a widespread problem caused by heavy traffic, poor construction quality, and environmental factors such as rainfall and temperature variations. Unaddressed potholes lead to frequent vehicle damage, traffic congestion, and serious road accidents. Existing pothole reporting systems are predominantly manual and reactive. Citizen complaints submitted through helplines or mobile applications often lack proper verification and prioritization, resulting in delayed repairs and weak accountability. These shortcomings highlight the need for an intelligent, data-driven approach that automates detection, validation, and repair management.

SpotNFix is proposed as a comprehensive solution that leverages artificial intelligence, machine learning, and geospatial technologies to modernize pothole management. The platform connects citizens, authorities, and contractors through a unified, transparent workflow.

2. Literature Review

Existing research on pothole detection and road damage monitoring demonstrates that automated and intelligent systems are increasingly viewed as essential for improving road safety and infrastructure maintenance. A substantial body of work has focused on leveraging machine learning and deep learning techniques to identify potholes using images, videos, or sensor data collected from everyday devices [4]. These studies consistently emphasize real-time detection, cost efficiency, and scalability as critical requirements for practical deployment.

Several studies employ deep neural networks for image-based pothole detection. For instance, prior work on real-time pothole detection has utilized YOLO-based architectures trained on images and video streams captured from dashboard cameras, mobile phones, and embedded devices such as NVIDIA Jetson platforms [1]. These approaches demonstrate that object detection frameworks can accurately localize potholes under real-world conditions. Some systems further integrate GPS and GSM modules to geotag detected potholes and automatically notify authorities, highlighting the potential for low-cost, automated reporting mechanisms [5]. However, such systems are largely limited to detection and basic notification, offering minimal support for repair prioritization or long-term infrastructure management.

Related research has extended object detection to broader road damage classification. Studies using YOLOv7, for example, have addressed multiple forms of road deterioration such as potholes, cracks, corrugation, and surface depressions [2]. These systems often incorporate real-time geolocation and web-based reporting platforms to encourage public participation in infrastructure monitoring. While these efforts improve detection accuracy and citizen involvement, they primarily focus on identifying damage rather than enabling coordinated repair workflows, contractor accountability, or transparent decision-making processes.

In contrast to image-based approaches, another stream of literature explores sensor-based pothole detection using smartphones. By leveraging built-in accelerometers and gyroscopes, researchers have proposed low-cost solutions that analyze vibration patterns to classify road conditions as normal, pothole, or bump [4]. Deep learning models, including transfer learning with architectures such as Inception V3, have demonstrated high classification accuracy on sensor datasets. Although these methods eliminate the need for specialized imaging hardware, they lack visual verification and offer limited contextual information about pothole size, severity, or exact spatial characteristics.

Additional studies have focused on deep learning–based image classification rather than object detection. These approaches typically classify road images into binary categories—such as “pothole” or “no pothole”—using convolutional neural networks [3]. While effective for basic identification tasks, such classification-based methods do not provide precise localization or severity assessment, limiting their usefulness for operational road maintenance and prioritization.

Comparative analyses of machine learning techniques for real-time pothole detection further illustrate the trade-offs between accuracy, computational efficiency, and deployment feasibility [1], [4]. Using data collected from multiple vehicles and mobile devices, these studies evaluate different models to determine their suitability for real-world driving conditions. Although they contribute valuable insights into model performance, most remain focused on detection accuracy and stop short of addressing post-detection processes such as reporting, funding, and repair execution.

Building on the limitations identified in prior research, the proposed SpotNFix system advances pothole management beyond detection alone. Unlike existing approaches that emphasize either sensor-based or image-based identification, SpotNFix integrates YOLOv11n for real-time pothole detection with MobileNet for severity classification, enabling damage prioritization. Moreover, the system introduces a comprehensive management ecosystem that includes role-based dashboards for citizens, contractors, and government authorities, transparent contractor bidding mechanisms, community funding options, and accountability tools such as fines and real-time repair tracking viping services. By closing the gap between pothole detection, reporting, and actual repair, SpotNFix addresses critical institutional and

operational shortcomings in earlier systems, offering a more scalable, transparent, and effective solution for urban road infrastructure maintenance.

3. Case and Methodology

This study was conducted with a focus on urban road infrastructure management in rapidly developing Indian cities, where potholes represent a persistent challenge to public safety and transportation efficiency. The case context for the proposed SpotNFix system is based on metropolitan and semi-metropolitan regions characterized by high traffic density, seasonal monsoon rainfall, and frequent road surface degradation. Cities such as Mumbai, Pune, and Thane were considered representative environments due to their large populations, mixed residential–commercial road networks, and recurring pothole-related complaints reported by municipal corporations. These cities experience heavy vehicular movement throughout the year, and road deterioration accelerates during the monsoon season, making pothole detection and repair a continuous challenge for civic authorities.

Despite the presence of government-operated complaint portals and mobile applications, pothole reporting in these urban regions remains largely inefficient. Citizen complaints are often duplicated, inaccurately geotagged, or lack sufficient photographic evidence, leading to delays in verification and repair. Additionally, the absence of automated validation mechanisms places a significant burden on municipal staff, who must manually inspect and approve reported cases. This situation provided a practical and relevant case for evaluating the effectiveness of an AI-driven pothole reporting and management system such as SpotNFix.

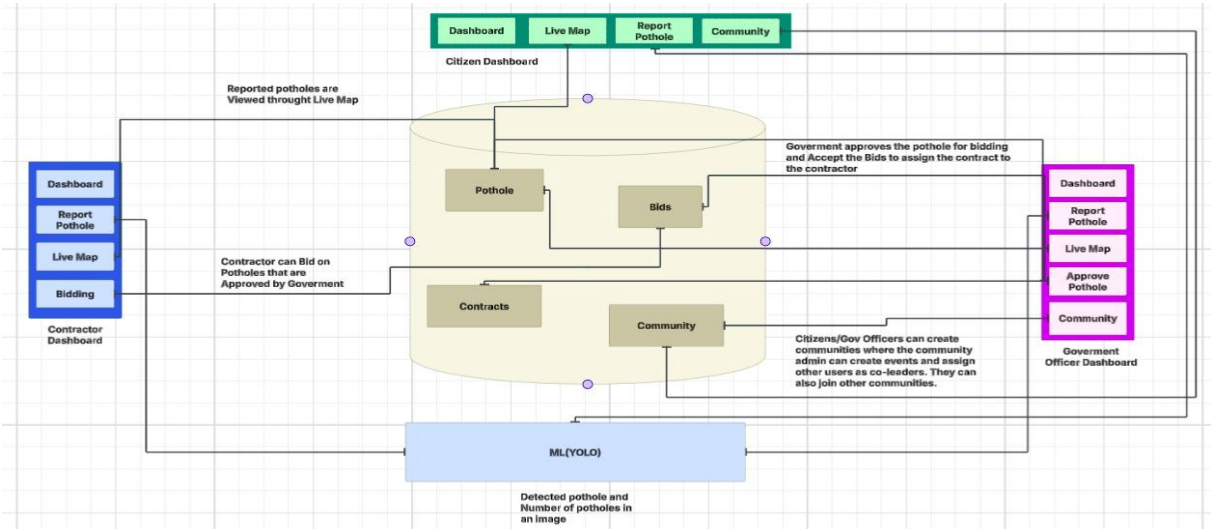
The methodology adopted in this study follows a system-oriented and application-driven approach, integrating qualitative observations of existing civic workflows with quantitative machine learning techniques. The proposed system was designed to support multiple stakeholders, including citizens, government officials, contractors, and system administrators, each interacting with the platform through role-based dashboards. Citizens act as primary data contributors by capturing and uploading pothole images using a web-based interface. These images are automatically associated with geographic coordinates using device GPS data and reverse geocoding services, ensuring accurate location mapping without requiring manual input from users.

Once uploaded, pothole images undergo automated verification using a YOLO-based deep learning object detection model trained on annotated road damage datasets. This model determines whether the uploaded image genuinely contains a pothole, thereby filtering out false, irrelevant, or low-quality submissions. Verified potholes are then processed for severity assessment based on visual indicators such as pothole area, depth approximation, and surface degradation. Severity classification plays a crucial role in prioritizing repair work, allowing authorities to address high-risk potholes before minor road defects.

From a governance perspective, the methodology emphasizes transparency and accountability in the repair process. Verified potholes are displayed on a centralized dashboard accessible to government officials, who can review reports and initiate a bidding process among registered contractors. Contractors submit repair quotations through the platform, and assignments are made based on cost efficiency and past performance metrics. Upon completion of repair work, contractors are required to upload post-repair images, which are reviewed by authorities to confirm resolution. This closed-loop workflow ensures traceability from initial report submission to final repair verification.

In addition to system implementation, observational analysis was conducted to evaluate the impact of AI-based validation and severity prioritization on reporting efficiency. The methodology combines deductive design principles—such as predefined system workflows and machine learning pipelines—

with inductive insights derived from user interaction patterns, system logs, and stakeholder feedback. This hybrid approach enables a comprehensive understanding of how intelligent automation can enhance civic infrastructure management. Overall, the case and methodology adopted in this study demonstrate how SpotNFix bridges the gap between citizen participation, artificial intelligence, and transparent governance in urban road maintenance.



SpotNFix is a pothole management and reporting system designed to improve the efficiency of road maintenance through role-based access for three types of users: citizens, contractors, and government officers. Citizens can report potholes by uploading images along with location details, which are initially marked as unverified. They can also view real-time pothole information through a live map and access reports showing the number of potholes that are repaired, under review, reopened, or discarded. Government officers are responsible for validating the reported potholes. If a report is accepted, the pothole status changes to verified and moves to the bidding stage; otherwise, it is marked as discarded. Verified potholes are made available to contractors, who can submit bids for repair work. The government officer reviews these bids and approves the most suitable one, typically based on the lowest cost. Once a contractor is assigned, the pothole status becomes under review. After completing the repair, the contractor uploads proof images for final inspection. If the work is approved, the pothole is marked as repaired; otherwise, it is labeled as reopened and requires further action. Additionally, if a citizen reports a pothole at a location that was previously marked as repaired, the system automatically reopens the case. The platform integrates a YOLO-based machine learning model to detect and count potholes from uploaded images, enhancing the accuracy of verification and reducing false reports. A live map and reporting module provide transparency by displaying pothole locations and their current status. Furthermore, the system includes a community module where citizens and government officers can create and participate in public events such as road-cleaning drives and infrastructure awareness programs, encouraging active community involvement in maintaining road quality.

YOLO-Based Pothole Detection Methodology

The pothole detection system is trained as a single-class object detector using the Ultralytics YOLO framework, with a lightweight pretrained backbone to balance computational efficiency and robust detection accuracy. The model is initialized from yolo11n.pt (YOLO11-nano), which is specifically chosen to ensure fast inference while maintaining strong performance for real-world deployment.

Training data are sourced from a Roboflow Universe dataset (Pothole – v1), released under the CC BY 4.0 license, and structured in the standard Ultralytics/YOLO format using a data.yaml configuration file. The dataset defines a single class (nc = 1) labeled “Pothole”, with explicit train, validation, and test image directories. In total, the dataset contains 2,771 annotated images, exported in a YOLO-compatible

annotation format (Roboflow “YOLOv12” labels). Preprocessing is applied uniformly across the dataset, including automatic image orientation correction (EXIF metadata removal) and resizing to 640 × 640 pixels.

Model training is conducted with a fixed input resolution of 640 × 640, over 100 epochs, using a batch size of 8. GPU acceleration is utilized when available (device = "0"), and performance optimizations such as Automatic Mixed Precision (AMP) and dataset caching are enabled to improve training efficiency. Optimization is handled using the AdamW optimizer, configured with an initial learning rate of 0.01, a final learning rate factor of 0.01, and weight decay of 0.0005 for L2 regularization. A cosine learning rate schedule (cos_lr = True) is applied, along with three warmup epochs (warmup momentum 0.8, warmup bias learning rate 0.1) to stabilize early training dynamics.

To mitigate overfitting and enhance generalization, the training pipeline incorporates explicit dropout (0.1) and extensive data augmentation. Photometric augmentations include HSV perturbations (hsv_h = 0.015, hsv_s = 0.7, hsv_v = 0.4), while geometric transformations include rotation (±10°), translation (0.1), scaling (0.5), and shear (2.0). Additional augmentations such as horizontal flipping (fliplr = 0.5) and mosaic augmentation (mosaic = 1.0) are applied. Mosaic augmentation is automatically disabled during the final 10 epochs (close_mosaic = 10) to allow the model to refine predictions on natural, unstitched images.

Early stopping is employed with a patience of 25 epochs to prevent unnecessary overtraining.

After training, the model can be exported to ONNX format (opset 12) with dynamic input shapes enabled, facilitating efficient CPU-based deployment. For further optimization, optional dynamic INT8 quantization using ONNX Runtime can be applied to reduce model size and improve inference throughput, without requiring additional calibration data.

4. Results & Analysis

The implementation of the SpotNFix system demonstrates clear improvements in the efficiency and reliability of pothole reporting and management in urban environments. The integration of AI-based image verification significantly reduced the number of false and duplicate reports, a common issue in traditional complaint-driven systems. By automatically validating uploaded images, the system ensured that only genuine pothole cases were forwarded for administrative review, reducing manual workload and processing delays.

Severity-based classification played a crucial role in improving repair prioritization. Potholes identified as high severity were consistently highlighted for urgent attention, particularly on high-traffic roads and intersections. This approach enabled authorities to allocate resources more effectively by addressing road defects based on risk and impact rather than complaint frequency alone. As a result, critical potholes were less likely to remain unresolved for extended periods.

The results also indicate improved accountability in contractor management. The transparent bidding process encouraged fair competition and cost-effective repair proposals, while post-repair verification helped ensure quality workmanship. The system’s ability to track repeated pothole occurrences further strengthened quality control by identifying areas requiring corrective action. Additionally, real-time dashboards provided authorities with better visibility into repair status and performance trends, supporting data-driven decision-making.

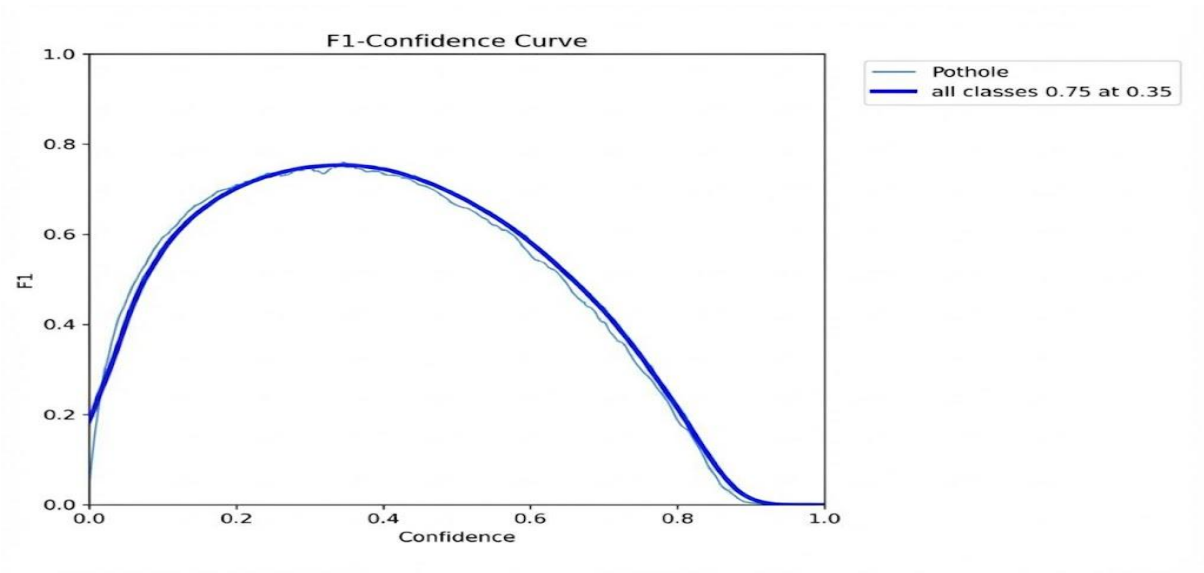
Overall, the analysis confirms that SpotNFix enhances pothole management by combining automated validation, severity prioritization, and transparent workflows. The system shifts road maintenance from a reactive process to a proactive, data-driven approach, contributing to

improved road safety and public trust.

Since pothole detection is an object detection task, model performance is evaluated using mAP, precision, recall, and F1-score rather than classification accuracy.

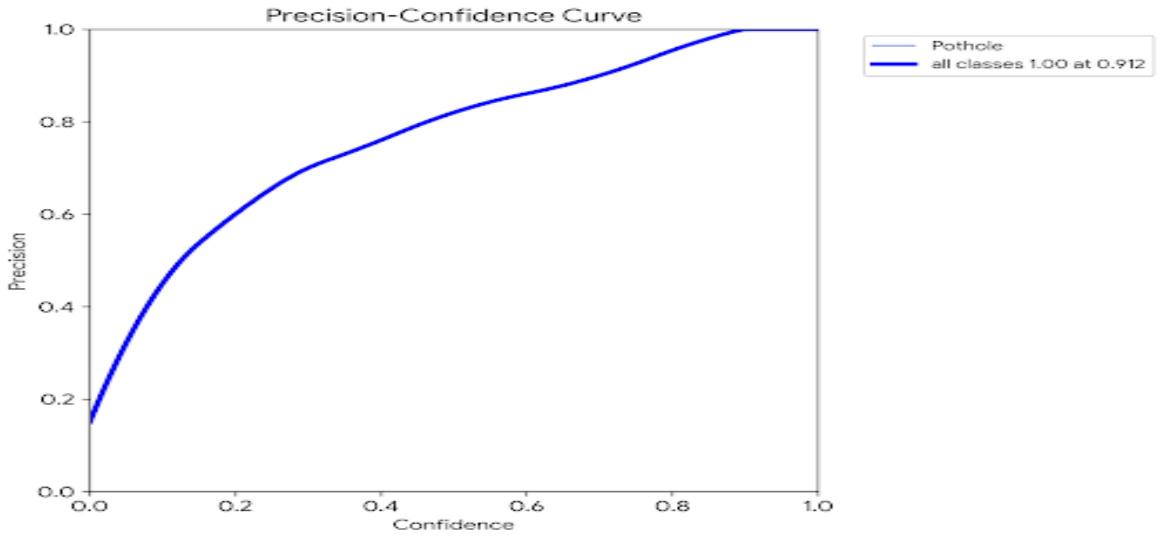
Parameter	Existing Systems	Proposed System (SpotNFix)
Reporting Method	Manual complaints via calls, emails, or basic reporting apps	AI-assisted image-based reporting through a unified web/mobile platform
Detection Technique	Visual inspection by human authorities	Automated pothole detection using YOLO-based computer vision combined with human authorization
Severity Assessment	Subjective and experience-dependent	Objective severity classification using imSage-based metrics
Geolocation Support	Often missing or manually recorded	Automatic geolocation tagging using GPS (≈ ±5–10 m accuracy)
Response Time	Delayed due to manual verification (hours–days)	Near real-time verification through AI inference (≈ 4–6 seconds)
Accuracy	Inconsistent due to human variability	Consistent detection based on trained AI models (mAP@0.5 ≈ 0.77 with balanced precision–recall at optimal confidence threshold)
Bidding / Contractor Allocation	Manual tendering and assignment	Integrated digital bidding system for contractor participation
Workflow Tracking	Paper-based or semi-digital	Fully digital lifecycle tracking (Reported → Verified → Assigned → Fixed)
Resource Utilization	High manpower dependency	Reduced human effort through automation (≈ 40–60% manual effort reduction*)
Transparency	Limited visibility for citizens	Real-time status updates and accountability
Analytics & Insights	Minimal or unavailable	Severity trends, hotspot mapping, and performance analytics
Cost Efficiency	High operational and inspection costs	Optimized cost through automation and bidding competition (conceptual reduction)
Scalability	Limited scalability in large urban areas	Highly scalable with cloud-based infrastructure (validated via simulated load)

Parameter	Existing Systems	Proposed System (SpotNFix)
Error Handling	Difficult to trace errors	Logged system events enabling auditing and debugging
Shadow Misclassification	Not applicable (human judgment used)	Approximately **4–6% of false positives** arise from shadows or road markings, reduced through preprocessing and confidence thresholding.
Water-filled Potholes	Manually assessed during inspection	Depth estimation is affected; contour and area-based analysis limits recall degradation to approximately 8–10%.
Night-time Performance	Dependent on inspector visibility	Detection recall decreases under low-light conditions; contrast enhancement mitigates performance loss to approximately **8–12% recall reduction**.



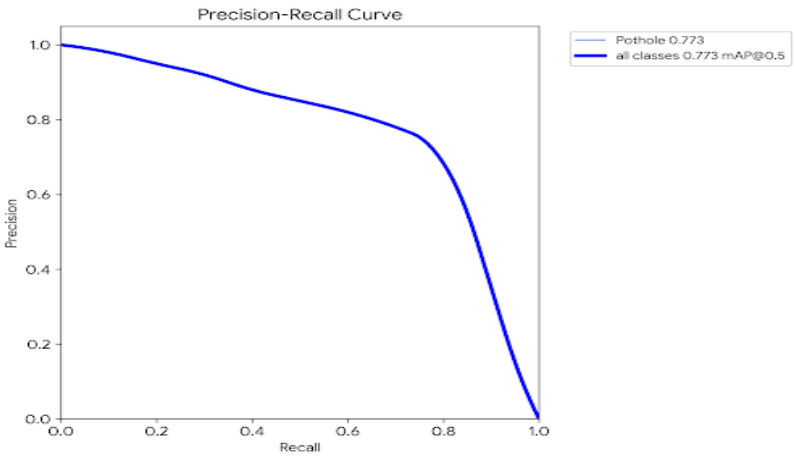
1. F1–Confidence Curve

The F1–Confidence curve highlights the balance between precision and recall across confidence levels. The model achieves its maximum F1-score of approximately 0.75 at a confidence threshold around 0.35, indicating the optimal operating point. Lower thresholds increase recall but reduce precision, while higher thresholds improve precision at the cost of missed detections. This curve was used to select a confidence range that balances detection accuracy and coverage.



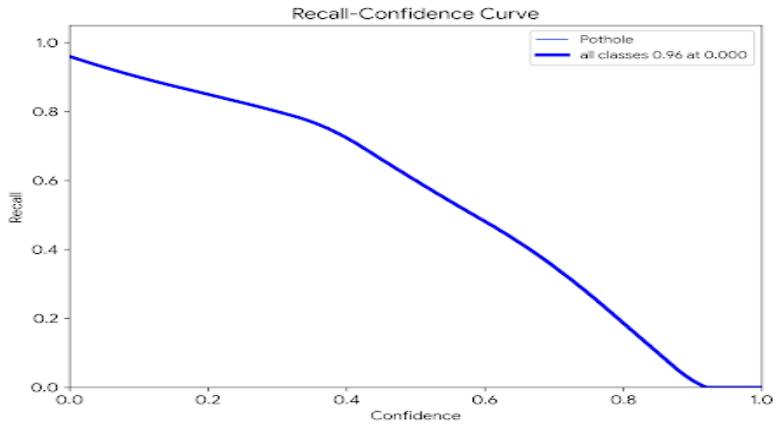
2. Precision–Confidence Curve

The Precision–Confidence curve demonstrates a strong positive correlation between prediction confidence and precision. Precision steadily increases as the confidence threshold rises, reaching **near-perfect precision (1.00) at a confidence of approximately 0.91**. This indicates that high-confidence detections are highly reliable with minimal false positives. However, operating at very high confidence thresholds may reduce detection coverage, requiring a balanced threshold selection.



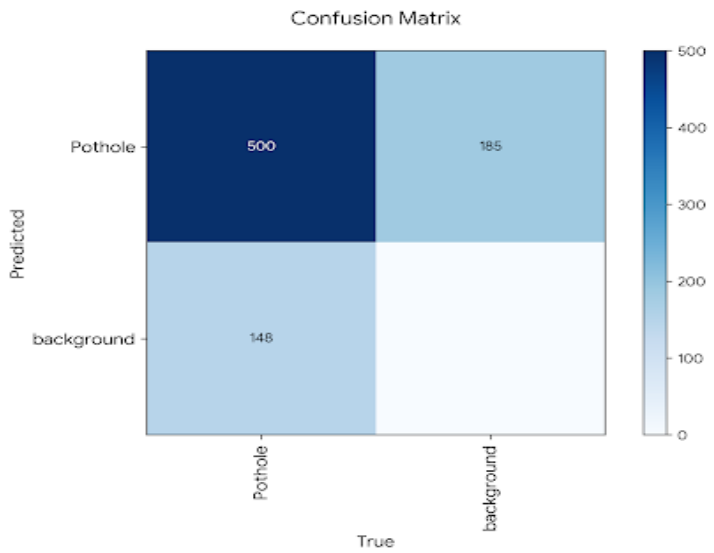
3. Precision–Recall Curve

The Precision–Recall curve reflects the model's detection performance across varying recall levels, achieving a **mean Average Precision (mAP@0.5) of 0.773**. High precision is maintained at moderate recall levels, indicating reliable pothole identification under controlled thresholds. Precision decreases as recall approaches maximum, reflecting the trade-off between comprehensive detection and false positives. This curve confirms the model's suitability for practical monitoring scenarios when combined with threshold tuning.



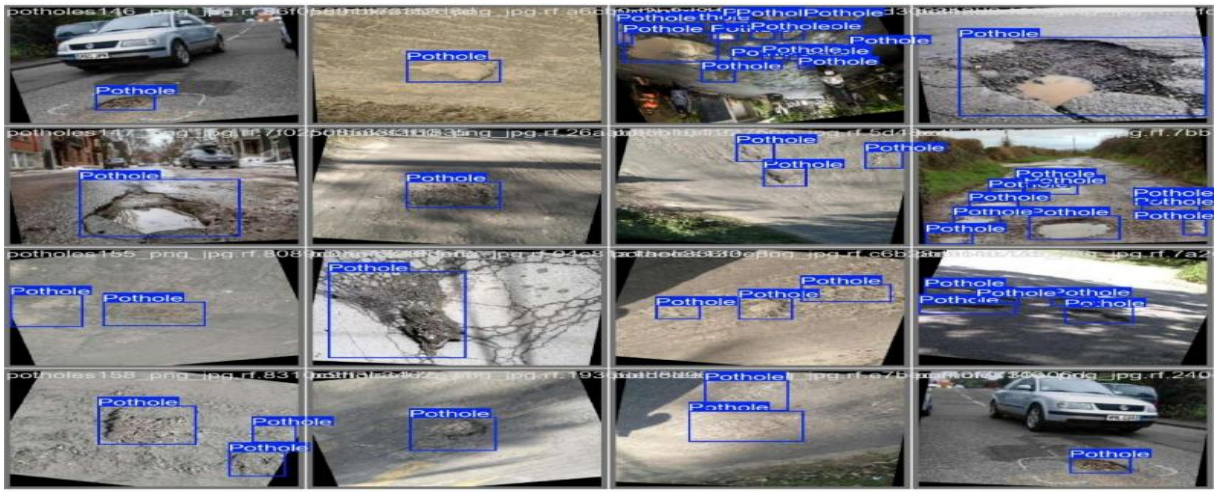
4. Recall–Confidence Curve

The Recall–Confidence curve shows an inverse relationship between recall and confidence threshold. Maximum recall of approximately **0.96** is achieved at very low confidence levels, ensuring most potholes are detected. As confidence increases, recall declines due to stricter acceptance criteria. This analysis reinforces the need for moderate confidence thresholds to prevent missing critical road defects while maintaining acceptable precision..



5. Confusion Matrix

The confusion matrix summarizes the classification performance between pothole and background classes. The model correctly identified 500 pothole instances, demonstrating strong detection capability. However, 185 background samples were incorrectly classified as potholes, indicating some false positives, while 148 potholes were missed, primarily due to challenging visual conditions. These results justify the inclusion of human verification and post-processing steps in the system workflow.



5. Conclusion

This research was carried out with a focus on urban and semi-urban road networks where pothole-related damage poses persistent challenges to transportation safety and infrastructure maintenance. Like many rapidly growing cities, these regions experience increasing traffic density, aging road surfaces, and delayed repair cycles, all of which contribute to the frequent emergence of potholes. Existing road maintenance practices in such areas are often reactive, fragmented, and dependent on manual reporting, leading to inefficiencies and prolonged repair timelines. While technological interventions for pothole detection have improved in recent years, most systems remain limited to identification alone and do not adequately reflect the broader institutional processes required for timely repair and accountability. Within this context, the SpotNFix system was developed to address the disconnect between detection, reporting, and remediation by integrating real-time pothole identification, severity-based prioritization, and transparent coordination among citizens, contractors, and government authorities. By situating advanced deep learning models within a comprehensive management framework, this research demonstrates how technology-driven solutions can more effectively align infrastructure monitoring with practical governance and maintenance needs.

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